

**INTEGRATION OF SAR AND OPTICAL
SATELLITE DATA FOR DELINEATION OF
DRY AND WET SNOW COVER IN THE
HOHE TAUERN NATIONAL PARK,
AUSTRIA**

MASTER'S THESIS

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ABSTRACT

Seasonal snow cover is a vital component of the alpine cryosphere, regulating mountain hydrology and reflecting global climate change impacts through shifting accumulation and melting patterns. However, accurate snow monitoring in high-relief environments like the Hohe Tauern National Park is frequently hindered by persistent cloud cover and the complex spatial variability of snow states. This thesis presents a synergistic approach for the high-resolution delineation of wet and dry snow by integrating Sentinel-1 Synthetic Aperture Radar (SAR) and Sentinel-2 optical satellite data.

The research utilizes a cloud-based workflow implemented in Google Earth Engine (GEE) to process multi-year time series from 2018 to 2025. Total snow-covered area is mapped using the Normalized Difference Snow Index (NDSI) from Sentinel-2, while the presence of liquid water in the snowpack (wet snow) is identified through C-band backscatter change detection from Sentinel-1. By fusing these datasets, the study successfully discriminates between Non-Snow, Dry Snow, and Wet Snow states at sub-kilometer resolution. The classification performance is validated using high-elevation station data and Receiver Operating Characteristic (ROC) analysis, demonstrating the robustness of multi-sensor fusion over single-sensor approaches.

The findings highlight the spatiotemporal progression of snowmelt across elevation gradients and its correlation with 'Alpine Warming' trends. The developed framework provides a reproducible and scalable tool for cryosphere monitoring, supporting water resource management and glacier conservation strategies in the Austrian Alps.

KEYWORDS

Sentinel-1, Sentinel-2, Normalized Difference Snow Index, Principal Component Analysis, Google Earth Engine, C-band backscatter, alpine cryosphere, elevation-dependent variability, NDSI, Snow Classification (Wet Snow/Dry Snow)

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DECLARATION

This thesis has been composed by *Syeda Noor ul Saba Bukhari* for the Erasmus Mundus Joint Master's Degree Program in Copernicus Master in Digital Earth for the academic years 2022/2023 and 2023/2024 at the Department of Geoinformatics, Faculty of Natural Sciences, Paris Lodron University Salzburg, and Department of Geoinformatics, Faculty of Science, Palacký University Olomouc.

Hereby, I declare that this piece of work is entirely my own, the references cited have been acknowledged and the thesis has not been previously submitted to the fulfilment of the higher degree.



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Theses guidelines

The aim of the thesis is to develop a high-resolution framework for wet and dry snow monitoring in the Western Austrian Alps by integrating Sentinel-1 SAR data with Landsat/Sentinel-2 optical imagery. By combining Sentinel-1 backscatter ratios for wet-snow identification with optical snow-extent mapping, the workflow overcomes the limitations of single-sensor approaches, such as cloud contamination in optical data and the difficulty of distinguishing snow states using SAR alone. The resulting multi-sensor analysis for the 2024–2025 period will provide more accurate spatial and temporal information on snow distribution, supporting protection management and hydrological risk assessment. A further aim is to design an effective visualisation workflow to present and communicate the derived snow products. The student will create a website about the thesis following the rules available on the department's website and a poster about the diploma thesis in A2 format. The student will submit the entire text (text, attachments, poster, outputs, input and output data) in digital form on a storage medium to the department's secretary.

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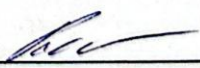
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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AUC	Area Under the Curve
DEM	Digital Elevation Model
ESA	European Space Agency
GEE	Google Earth Engine
GIS	Geographic Information System
GRD	Ground Range Detected
IW	Interferometric Wide swath
LST	Land Surface Temperature
MODIS	Moderate-resolution Imaging Spectroradiometer
NDSI	Normalized Difference Snow Index
NDVI	Normalized Difference Vegetation Index
PCA	Principal Component Analysis

INTRODUCTION

Alpine snow cover plays an essential role in the water cycle of central Europe. It can be seen as a vital component for the seasonal water yield, as well as the groundwater recharge, and the freshwater supply for numerous people living downstream. In Austria, the snow cover time series from profiling stations shows a significant decrease in snow depth and winter accumulation, particularly during the core winter (Schöner et al., 2009). Based on model calculations Olefs et al. (2020) found 42 fewer snow-covered days and a reduced seasonal snow depth by 0.15m in Austria between 1961/62 and 2019/20. Moreover, the strongest reductions were found at low elevation below 1,500 m a.s.l. The model results further suggest that snow-covered area in high-altitude catchments in the Austrian Alps decreased throughout the 21st century, which will impact seasonal runoff regimes (Hanzer et al., 2018). Such information is of particular interest for water resource management as it helps understand better how alpine snow cover influences water supply and quality.

At the same time, the snow cover in mountains can be challenging to observe because of the complexity of the terrain. Remote sensing is the only way to provide consistent spatial and temporal information about snow cover in hard-to-reach areas such as the alpine region. While the method proved its effectiveness in monitoring the snow cover extent with daily resolution using optical satellite observations in Austria (Tong et al., 2020), the authors highlight the limitations due to the coarse resolution. Tong et al. (2020) further observe that using higher-resolution data, such as Sentinel-2, allows mapping snow cover at higher accuracy. This is accomplished by determining the normalized difference snow index (NDSI) and using thresholds that take into account the variability of snow-free surfaces with different aspects and land cover types and the timing of snow cover duration (Ma & Zhang, 2022). Another limitation of current optical methods is connected to the frequent cloud cover blocking the view of the ground. On the other hand, Synthetic Aperture Radar (SAR) can ensure the continuous monitoring of snow cover throughout the year. C-band SAR is sensitive to liquid water in snow; Brangers et al. (2024) note that the backscatter of snow cover can be predicted based on the microstructure and layering of the snow surface at the time of measurement. Lund et al. (2022) further suggest that observations can be used in combination with field measurements and snowpack models to discriminate effectively between wet and dry phases throughout the entire snow season the present research will therefore attempt to design and test the concept for snow cover monitoring based on the combined observations from Sentinel-1 and Sentinel-2. The new algorithm was tested in Hohe Tauern National Park in Austria between 2018 and 2025. The study uses Principal Component Analysis to combine a composite of eight bands of optical reflectance and SAR backscatter measurements. The component is used to separate pixels of snow-covered and snow-free land cover. Further, the classification follows the hierarchical snow pixel-based approach and groups dry and wet snow phases at different altitudinal zones (three classes: Non-Snow, Dry Snow, and Wet Snow). The analysis attempts to use the advantages of both optical and SAR methods to overcome their limitations and provide a consistent observation framework for long-term snow cover monitoring.

1 OBJECTIVES

The study aims at designing a multi-sensor remote sensing method to support the identification and classification of snow cover using Sentinel-1 SAR and Sentinel-2 optical imagery over Hohe Tauern National Park, Austria, for the period 2018–2025. In particular, this research will develop a PCA-based approach for the fusion of multi-sensor data and propose a set of rules for hierarchical classification of non-snow, dry snow, and wet snow cover to assess their relationships with climatic variables for the four elevation classes.

The main objectives of the thesis are:

1. To integrate Sentinel-1 SAR and Sentinel-2 optical data using Principal Component Analysis for improved snow cover detection across Hohe Tauern National Park.
2. To classify dry and wet snow states by combining Sentinel-2-derived spectral indices with SAR backscatter thresholds.
3. Explore relationships between snow states and climate variables (precipitation and temperature), and snowmelt and wet and dry snow

2 STATE OF ART

This chapter presents an overview of the published scientific knowledge and previous research on remote sensing of seasonal snow cover. It provides the theoretical foundations of optical spectral indices and SAR backscatter observations for the assessment of total snow extent, as well as the identification of liquid water content in snow covers. The current state of the art in multi-sensor data fusion is discussed, while the climatic characteristics of the Austrian Alps and Hohe Tauern National Park are considered for the regional analysis of snow conditions. Altogether, this chapter sets the scene for the following study that addresses the scientific gap in the high-resolution delineation of wet and dry snow in mountainous areas.

2.1 Snow cover remote sensing and spectral indices

Optical satellite remote sensing has been the dominant approach for large-scale snow cover mapping since the early 1990s. The normalized difference snow index (NDSI) is calculated using surface reflectance in the visible and shortwave-infrared spectral bands. The high reflectance of snow in the green band and its strong absorption at 1.6 μm wavelength allow distinguishing snow from most other land covers. The NDSI value ≥ 0.4 is generally accepted as a threshold for pixel-specific binary snow classification (Poussin et al., 2023). For heterogeneous mountainous areas, where the accuracy of snow mapping is challenging due to extensive shadows, mixed pixels, and forested regions, an improved algorithm of automated threshold selection was suggested by Ma & Zhang (2022). Their multi-factor dynamic NDSI threshold proved its efficiency in delineating snow from forests and shaded areas in a high-relief landscape. Tong et al. (2020), in their study on the hydrological modeling of an alpine catchment in Austria, demonstrated that the assimilation of daily MODIS Collection 6 snow cover products significantly improved the simulation of streamflow dynamics. The algorithm used the spatially continuous snow cover information to adjust model parameters that were poorly captured by conventional remote sensing data.

2.1.1 SAR backscatter and snow detection

The use of Synthetic Aperture Radar (SAR) imaging is partially complementary to the information derived from optical sensors since, unlike optical data, SAR is independent of solar irradiance conditions. This capability is especially useful for monitoring mountainous snowpacks, which can be frequently affected by cloud cover, particularly during the spring period when the melt season begins. The underlying physical mechanism that enables wet snow detection with SAR is related to the change in dielectric properties of snowpack with a transition from solid to liquid state. Specifically, liquid water content significantly increases snowpack's dielectric constant, thereby decreasing C-band backscatter by several dB as compared to an ice-dominated or bare soil reference. The first study to utilize this principle for operational wet snow detection in the European Alps was conducted by Nagler et al. (2016), who applied a change detection algorithm to VV-polarization of Copernicus Sentinel-1 SAR data, thereby enabling automated delineation of the snowmelt front. This method was later found to be applicable to a variety of terrain and vegetation through the regional analysis performed by Tsai et al. (2019). Notably, these researchers highlighted that detecting dry snow with SAR is inherently difficult due to its low liquid water content which renders it almost invisible to microwave radiation. The same limitation was addressed by Marin et al. (2020) in their study of the

Italian Alps, where a temporal stack of Sentinel-1 data combined with derivative topographic information and land cover classification enabled consistent detection of snowpack melt from across the study area. At a more fundamental level, Brangers et al. (2024) analysed C-band radar backscatter from a tall tower to characterise the sensitivity of SAR observations to liquid water content in the snowpack. Finally, Carletti et al. (2025) analysed multitemporal Sentinel-1 acquisitions during the snowmelt period in conjunction with high-resolution field measurements and radiative transfer modelling to identify changes in backscatter characteristics associated with changes in snow physical properties.

2.1.2 Dry and wet snow classification

The ability to discriminate between dry and wet snow greatly improves the utility of snow monitoring products for water management and avalanche forecasting applications, although the snow classification is not straightforward. Varade et al., (2019) showed that using the combination of dual-polarisation SAR observations with multispectral indices can allow for the separation of dry and wet snow classes in high-mountainous regions. In particular, the authors highlight the benefit of using SAR information, which contains the phase-state information, unavailable in optical bands, for improving the discrimination of the snow class. Karbou et al. (2021) highlights the potential of using Sentinel-1 backscatter time-series together with numerical weather prediction data for monitoring seasonal wet-snow events and distinguishing between melt-affected and dry snow or snow-free areas at weekly timescales. Lund et al., (2022) employed Sentinel-1 observations in combination with a physical snow model SnowModel to relate backscatter variations to the evolution of the snowpack surface during melt, freeze, and ripening periods.

2.1.3 Multi-sensor data fusion (SAR + optical)

The combined use of synthetic aperture radar (SAR) and optical remote sensing data has been identified as the most operationally efficient approach to snow cover monitoring through cloud cover. The SAR satellite Sentinel-1 and the optical sensor Sentinel-2 offer the best potential for combined snow cover mapping as both sensors have a comparable revisit capacity and spatial resolution. Therefore, cloud-free optical satellite images can be completed with SAR information on snow state and vice versa. Tsai et al. (2019) and Carletti et al. (2025) describe similar approaches that process combined SAR-optical features in order to retrieve improved snow extent and snow phase information. The present thesis follows this concept and applies a Principal Component Analysis (PCA) using Sentinel-1 backscatter coefficients (VV, VH), the spectral bands B2, B3, B4, B8 and B11 of Sentinel-2 as well as the Normalized Difference Vegetation Index as input features. The resulting components are applied to discriminate between areas of Non-Snow, Dry Snow and Wet Snow. In contrast to the approaches mentioned above, NDSI is not used as an input feature or a classification threshold.

2.1.4 Snow cover dynamics in the Austrian Alps

The Austrian Alps are among the European mountain regions expected to be most strongly impacted by snow cover trends. A significant tendency towards reduction of snow depth and snow season was identified for the high-altitude weather station Sonnblick, which is representative for the whole area (Schöner et al., 2009). Recently become available high-resolution gridded data sets of daily temperature and precipitation over Austria (1 km resolution, SPARTACUS: 1961-present) enabled to analyze the trends in

detail for various regions within the country (Hiebl & Frei, 2018). Using a model-based reconstruction approach, Olefs et al. (2020) assessed changes in snow depth, snow cover duration, and snow making potential for Austria in the period 1961-2020. For the same region, Hanzer et al. (2018) used a physically based cryosphere model to estimate the future snow pack trends under various climate scenarios. Their results demonstrate the strongest impact on the snow cover in higher altitudes. At a larger scale, Haslinger et al. (2023) related the snow pack trends to the drought development in the Eastern Alps, highlighting the societal impacts of reduced snowmelt contribution to the water resources. Hohe Tauern National Park is located in the highest and most glaciated part of the Alps in Austria. This area is characterized by a large altitudinal gradient and can be considered as a domain with complex topography. These properties make it especially interesting for the current study aimed at validating multi-sensor snow classifications.

3 METHODOLOGY

3.1 Study Area

This chapter introduces the study area, research methodology used in the study that is focused on analyzing the spatio-temporal patterns of seasonal snow cover in the Hohe Tauern National Park in the period 2018–2025. First, the methodology section briefly outlines the main steps of the current research workflow. Following the introduction, the next section thoroughly describes data preparation steps and discusses the details of SAR and optical image processing in Google Earth Engine. Additionally, the present section explains the principles of Principal Component Analysis, hierarchical rule-based algorithm, and statistical methods used in the current study to analyze the elevation trends and classify Non-Snow, Dry Snow, and Wet Snow. Finally, it should be noted that the methodology describes the procedures used to assess the performance of the proposed classification algorithm using the AUC–ROC statistical method.

3.1.1 Geographic Setting

Hohe Tauern National Park is situated in the central part of the Eastern Alps in Austria, covering the federal states of Carinthia (Kärnten), Salzburg, and Tyrol (Tirol). The area covered by the park is 1,856 km², which makes it the largest national park in the Alps and one of the largest protected natural areas in Central Europe (Hohe Tauern National Park: Valleys and Peaks in the Hohe Tauern National Park [online]. [Cit. 2026-07-16]. Available online: <https://www.nationalpark.at/en/nationalpark/valleys-and-peaks>). The national park is distributed along and on both sides of the main Alpine ridge, encompassing the area from the valley bottoms at approximately 700 m. a.s.l. up to the peak of the Großglockner, the highest mountain in Austria, at 3,798 m. a.s.l. The park's status as a protected area and the fact that most of it remains untouched by human activity make it a perfect site for studying natural snow cover. Since there are no anthropogenic sources of land cover change in the park, they should be excluded as a confounding factor from further analysis.

3.1.2 Topography

The topography of Hohe Tauern National Park is defined by its high and steep reliefs formed under the influence of Pleistocene glaciers. The territory of the national park covers one of the largest tectonic windows in the Alps – the Tauern Window, which includes a variety of metamorphic rocks of the Penninic nappe. As a result, the area has a relatively complex topography with significant slopes and large basins formed by glacial erosion. Many peaks in the district rise to more than 3000 meters above sea level, and hundreds of glaciers cover the area. In addition, the topography of Hohe Tauern National Park has a significant impact on the amount of solar radiation coming to the site, as well as the windward and leeward sides of the mountains. Consequently, the distribution of snowline and snow belts can be affected by local winds and topography.

The digital topographic model used is a slope layer derived from the ALOS PALSAR radar satellite's 12.5 m resolution. Additionally, slope and aspect were calculated from this model to characterise the topography and general location of the study area. All data were projected to the WGS84 / UTM Zone 33N projection for the Sentinel-1 and Sentinel-2 satellites, which were used for research from 2018 to 2025.

3.1.3 Climate

The climate of Hohe Tauern National Park varies from oceanic in the west and north to more continental in the inner-alpine east. It is strongly determined by altitude since annual mean temperatures are close to 0 °C at the timberline (1800-2000 m a.s.l.) with a large annual amplitude, characteristic of high elevations. Annual precipitation amounts to 2000 mm or more in exposed western and northern mountain tops, with a significant part falling as snow in the form of heavy snowstorms in winter (October-May).

Snow in Hohe Tauern National Park is a key factor controlling the hydrological regime, ecology, and energy balance. Alpine snow cover in the region usually starts to form in October and is maximum in February-March, being gradually depleted in spring and summer. The properties of the snow cover, including its duration, date of maximum snow depth, and snow water equivalent, may substantially vary from year to year and from one snow season to another. In the Austrian Alps, snow depth and snow cover duration decreased significantly at low and mid-altitudes during the last decades (Schöner et al., 2009; Olefs et al., 2020). However, the trends at high altitudes were found to be dependent on temperature and precipitation variability (Hanzer et al., 2018). The melting of snow cover in high-altitude regions is one of the main sources of river runoff in Austria, particularly for the Salzach, Drau, and Inn catchments, which are partly located in Hohe Tauern National Park (Hanzer et al., 2018).

3.1.4 Elevation Zones

Due to the strong dependence of snow accumulation, ablation, and snow state (wet/dry) on altitude, the study area was stratified according to elevation using the ALOS PALSAR Digital Elevation Model (DEM). The following four elevation zones were delineated:

Zone I (<1,500 m): the lower valley and mountainous area with discontinuous and seasonal snow cover and a relatively short snow season;

Zone II (1,500–2,000 m): the lower sub-alpine area with a stable seasonal snowpack and a distinct spring snowmelt peak identified in both optical and SAR data;

Zone III (2,000–3,000 m): the alpine area with deep and prolonged snowpack dominating the snow cover extent during the cold season;

Zone IV (>3,000 m): the high-alpine and glacierized area with a continuous and perennial snow cover and the largest challenge for optical remote sensing due to frequent cloudiness and low sun angles during the winter season.

The stratification scheme allows assessing the relationships between elevation and snow cover properties and evaluating differences in the classifier's performance and sensor's capabilities for different snow regimes. The spatial distribution of the stratified elevation zones is presented in Fig. 3.1.

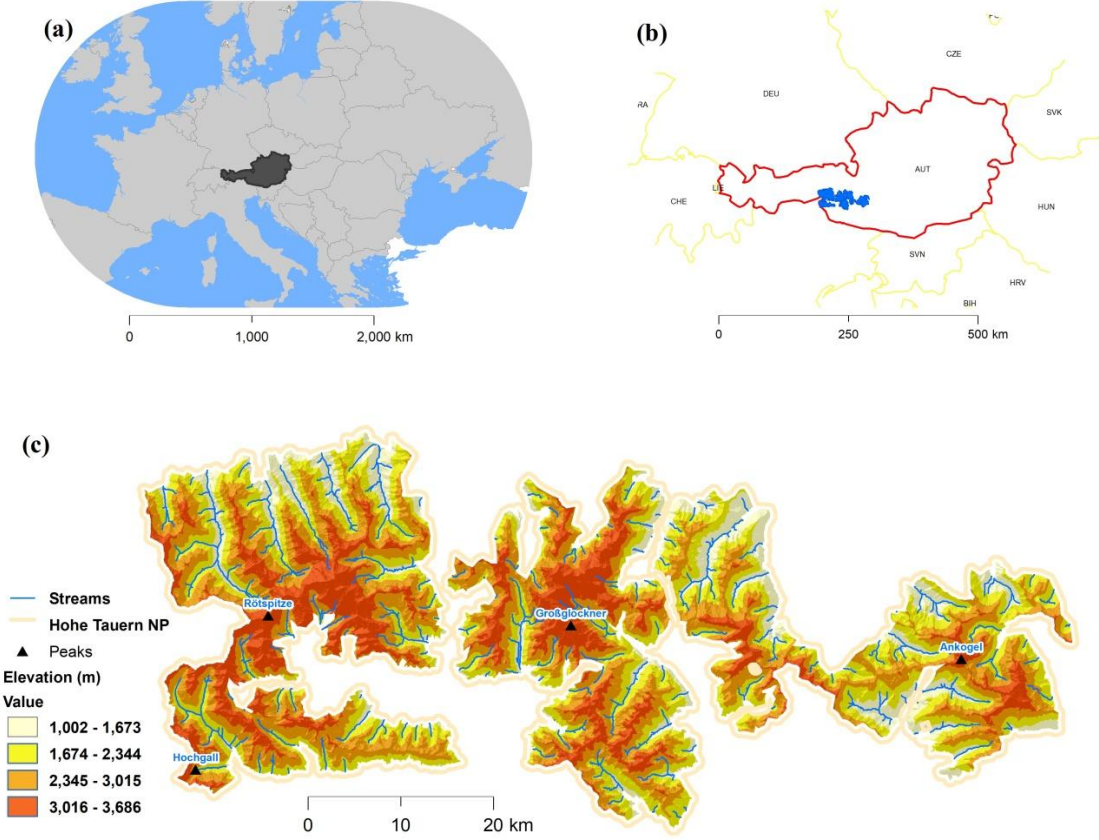


Figure 1 Location and topography of Hohe Tauern National Park: (a) location of Austria within Europe, (b) location of Hohe Tauern National Park within Austria, and (c) elevation, streams, and selected peaks within the national park.

3.2 Overall Research Framework

The study proposes an integrated and innovative approach used for the detection, identification, and analysis of temporal tendencies of snow cover in the High Tauern National Park, Austria, in 2018–2025 (Fig. 2). It includes various methods used for multi-sensor and multi-source data analysis, such as multi-spectral optical data from the Sentinel-2 satellites, Synthetic Aperture Radar (SAR) backscatter from Sentinel-1, topographical data in the form of a Digital Elevation Model (DEM), and gridded climate data from the SPARTACUS reanalysis product. The general process consisted of five stages: (i) pre-processing of the input data and calculation of relevant spectral indices; (ii) analysis of the combined data from Sentinel-2 and Sentinel-1 using Principal Component Analysis (PCA). The results were used to calculate the NDSI; (iii) creation of a rule-based algorithm for the classification of Non-Snow, Dry Snow, and Wet Snow; (iv) analysis of the elevation-based, seasonal, and climatic trends in snow cover; and finally, accuracy assessment using the AUC–ROC metrics.

3.2.1 Data Collection and Description

The main variables used as input data are presented in Table 1. The Sentinel-2 Level-2A (L2A) multispectral surface reflectance and Sentinel-1 Ground Range Detected (GRD) Synthetic Aperture Radar (SAR) data were downloaded from the Copernicus Open Access Hub and processed in Google Earth Engine (GEE). For each year of the study period (2018–2025), two seasonal composites were generated for each sensor. The first

composite covered the snow accumulation period (December–February), while the second one included the snow depletion period (June–September). Overall, 2025 is the last year of analysis, while 2018 is the first year of the considered period. Thus, for each sensor, 16 seasonal composites were generated. The number of images for the considered temporal window (15 June–15 September) varied between 44 (2022) and 117 (2018) images for Sentinel-1 and between 60 (2024) and 88 (2021) images for Sentinel-2. While for the snow accumulation period, the number of images for Sentinel-1 varied between 82 (2023–2025) and 220 (2018–2019) and for Sentinel-2, between 54 (2017–2018) and 149 (2021–2022) (Table 2).

The Sentinel-2 multispectral satellite data provided the spectral reflectance values in the visible (Blue (B2), Green (B3), Red (B4)), near-infrared (NIR, B8), and short-wave infrared (SWIR1, B11) domains for snow and vegetation indices estimation. For the Sentinel-1 SAR data, the C-band backscatter coefficients in VV and VH polarizations were used for the Interferometric Wide (IW) swath mode for the identification of frozen and melted snow cover. Additionally, the 12.5 m resolution ALOS PALSAR DEM data were used to estimate slope and aspect layers for altitude stratification. The SPARTACUS reanalysis data, provided by Geosphere Austria, were used as climate input data. The daily temperature and precipitation were extracted from the gridded data at 1 km resolution.

In addition, Normalized Difference Snow Index (NDSI) MODIS (500 m, Terra/Aqua satellites) and Landsat 8/9 (30 m, OLI sensor) data were used as independent data for the accuracy assessment. Similar to the main data, the Landsat data were retrieved from the GEE platform and processed to the Level-2 surface reflectance.

Table 1 Datasets used in the study, including sensor or source, data type, and spatial resolution.

Dataset	Sensor / Source	Data Type	Spatial Resolution
Sentinel-2 Level-2A (MSI)	Copernicus / ESA	Multispectral Surface Reflectance (B2, B3, B4, B8, B11)	10–20 m
Sentinel-1 GRD (SAR)	Copernicus / ESA	C-band SAR Backscatter (VV, VH polarizations, IW mode)	10 m
Digital Elevation Model (DEM)	ALOS PALSAR	Elevation, Slope, Aspect	12.5 m
Climate Reanalysis	SPARTACUS / Geosphere Austria	Temperature (°C), Precipitation (mm)	1 km
MODIS Snow Product	Terra/Aqua / NASA	NDSI-based Snow Cover	500 m

Dataset	Sensor / Source	Data Type	Spatial Resolution
Landsat 8/9 OLI	USGS / NASA	Multispectral Surface Reflectance	30 m

Table 2 Sentinel-1 and Sentinel-2 Satellite Images details for Year, date and number of images used.

Season	Year	Start Date	End Date	Sentinel-1 Image Count	Sentinel-2 Image Count
Depletion	2018	15-06-18	15-09-18	117	69
	2019	15-06-19	15-09-19	91	61
	2020	15-06-20	15-09-20	89	72
	2021	15-06-21	15-09-21	84	73
	2022	15-06-22	15-09-22	44	63
	2023	15-06-23	15-09-23	46	61
	2024	15-06-24	15-09-24	45	60
	2025	15-06-25	15-09-25	83	88
Accumulation	2018	15-12-17	28-02-18	164	54
	2019	15-12-18	28-02-19	220	113
	2020	15-12-19	28-02-20	174	102
	2021	15-12-20	28-02-21	168	102
	2022	15-12-21	28-02-22	119	149
	2023	15-12-22	28-02-23	82	79
	2024	15-12-23	28-02-24	83	84
	2025	15-12-24	28-02-25	82	128

3.2.2 Data Preprocessing

3.2.2.1 Sentinel-2 Optical Data Preprocessing

Sentinel-2 Level-2A imagery was accessed as atmospherically corrected surface reflectance products. Cloud-contaminated pixels were removed using the QA60 quality assurance band, which encodes cloud and cirrus flags at 60 m resolution resampled to the native pixel grid. Pixels flagged as cloud or cloud shadows were masked prior to compositing. Cloud-free seasonal composite images were generated using the median reducer within GEE for each defined seasonal window (accumulation: December–February; depletion: June–September) and each year from 2018 to 2025. Median compositing was preferred over mean compositing as it is more robust against residual outliers and undetected cloud contamination. All optical bands used in subsequent analysis (B2, B3, B4, B8, B11) were retained at their native 10–20 m resolution; bands at 20 m were resampled to 10 m using bilinear interpolation to achieve spatial consistency across the multi-spectral stack.

3.2.2.2 Sentinel-1 SAR Data Preprocessing

Sentinel-1 GRD products were filtered to include only Interferometric Wide (IW) swath mode acquisitions in ascending orbit configuration, ensuring consistency in incidence angle and acquisition geometry throughout the time series. Raw backscatter intensity values in linear power scale were converted to the logarithmic decibel (dB) scale using the standard transformation:

$$\sigma^{\circ} \text{ (dB)} = 10 \times \log_{10} (\sigma^{\circ} \text{ linear})$$

Seasonal median composites of VV and VH polarization backscatter were computed for each accumulation and depletion period to reduce the influence of speckle noise and to provide temporally stable backscatter estimates. Speckle reduction via median temporal compositing was selected in preference to spatial filtering to preserve spatial resolution. For the relative backscatter change analysis used in wet snow detection, a multi-year seasonal baseline (ΔVV reference) was computed as the median VV backscatter across all dry-season acquisitions within the study period, following established approaches in the SAR-based snow mapping literature (Lund et al., 2022; Carletti et al., 2025).

3.2.2.3 Multi-Sensor Co-registration and Normalization

All optical and SAR datasets were resampled to a uniform spatial resolution of 10 m and co-registered to a common spatial reference system (WGS84 / UTM Zone 33N) using GEE's reprojection functionality. Pixel-level alignment between Sentinel-2 and Sentinel-1 was verified through visual inspection and spatial overlay analysis. Prior to Principal Component Analysis (PCA), all input bands were subjected to mean-centering and z-score standardization independently to eliminate scale-related biases arising from the inherently different radiometric properties of optical reflectance and SAR backscatter data. Standardized values were computed per band as:

$$z = (x - \mu) / \sigma$$

where x is the pixel value, μ is the band mean, and σ is the standard deviation computed across all pixels within the Area of Interest (AOI). This normalization ensured equal weighting of all eight input variables in the subsequent PCA.

3.2.3 Spectral Index Derivation

Two spectral indices were derived from the Sentinel-2 surface reflectance composites to enhance thematic separability between snow, vegetation, and other land surface types. The Normalized Difference Snow Index (NDSI) was calculated as:

$$\text{NDSI} = (\text{B3} - \text{B11}) / (\text{B3} + \text{B11})$$

where B3 is the Green band (560 nm) and B11 is the Shortwave Infrared 1 (SWIR1) band (1610 nm). NDSI exploits the high reflectance of snow in the visible spectrum and its characteristically low reflectance in the SWIR region, enabling robust discrimination of snow-covered surfaces from other spectrally bright features such as clouds, sand, and salt flats. The Normalized Difference Vegetation Index (NDVI) was computed as:

$$\text{NDVI} = (\text{B8} - \text{B4}) / (\text{B8} + \text{B4})$$

where B8 is the Near-Infrared band (842 nm) and B4 is the Red band (665 nm). NDVI was incorporated into the multi-sensor stack and classification logic to minimize spectral confusion between snow and vegetated surfaces, particularly in areas of partial snow cover during the depletion season.

3.2.4 Principal Component Analysis and Information Content Assessment

To evaluate the integrated information content of the combined optical and SAR dataset, an eight-band input stack was assembled comprising five Sentinel-2 spectral bands (B2, B3, B4, B8, B11), NDVI, and two Sentinel-1 SAR backscatter bands (VV and VH). NDSI was deliberately excluded from the PCA input stack to avoid methodological circularity in the subsequent validation step, given that NDSI was used as an independent reference for snow cover in the correlation analysis. Following z-score normalization of all eight bands, Principal Component Analysis (PCA) was performed using the `getPrincipalComponents` function within GEE at a spatial resolution of 10 m. This orthogonal linear transformation decomposed the original correlated band space into a new set of uncorrelated Principal Components (PCs) ordered by decreasing proportion of explained variance.

The first eight principal components (PC1–PC8) were retained. To identify which component most effectively captured snow-related spectral variability, Pearson's correlation coefficient (r) was computed between each principal component and the independently derived Sentinel-2 NDSI layer across all pixels within the AOI. The component exhibiting the strongest correlation with NDSI was selected as the primary snow information carrier. Results of the correlation analysis were tabulated and visualized graphically to support interpretation of the spectral–radar data integration.

3.2.5 Rule-Based Snow Classification

A hierarchical rule-based classification algorithm was developed to categorize each pixel into one of three mutually exclusive classes: Non-Snow, Dry Snow or Wet Snow. The classification was executed in two sequential stages.

In the first stage, snow extent was delineated using an NDSI threshold approach. Pixels with $\text{NDSI} > 0.4$ were classified as snow-covered, while pixels with $\text{NDSI} \leq 0.4$ were assigned to the Non-Snow class. The threshold value of 0.4 was adopted based on established literature (Ma and Zhang, 2022; Poussin et al., 2023; Tong et al., 2020) and validated through visual comparison of classified outputs against high-resolution Sentinel-2 optical imagery. This threshold effectively separated snow from clouds, water bodies, and bright bare soil surfaces within the study area.

In the second stage, snow-covered pixels were further differentiated into Dry Snow and Wet Snow sub-classes using Sentinel-1 SAR backscatter. Wet snow, characterized by the presence of free liquid water within the snowpack, causes significant attenuation of C-band microwave signals due to increased dielectric losses, resulting in markedly reduced backscatter intensity (Brangers et al., 2024; Carletti et al., 2025; Lund et al., 2022). Two complementary criteria were applied to identify wet snow conditions: (i) an absolute

backscatter threshold, whereby pixels with VV backscatter < -11 dB were classified as Wet Snow; and (ii) a relative change criterion, whereby pixels exhibiting a backscatter decrease of $\Delta VV < -2.0$ dB relative to the dry-season baseline were classified as Wet Snow. Pixels satisfying either criterion were labeled as Wet Snow; all remaining NDSI-classified snow pixels were assigned to the Dry Snow class.

To reduce classification noise and improve spatial coherence, a minimum mapping unit (MMU) filter was applied post-classification. Isolated pixel clusters containing fewer than ten spatially connected pixels were removed from each snow class and reassigned to the dominant neighboring class. This step minimized the impact of residual speckle noise in SAR data and improved the visual and statistical consistency of class boundaries.

3.2.6 Land Surface Temperature (LST) Derivation

Land Surface Temperature (LST) was derived from Landsat 8/9 thermal infrared imagery (Band 10, TIRS) to provide a spatially distributed thermal context for interpreting snow cover variability across the study area. Seasonal mean LST composites were generated for each accumulation (December–February) and depletion (June–September) period from 2018 to 2025, yielding one mean LST map per season per year. LST retrieval was performed using the split-window algorithm or the single-channel method applied to atmospherically corrected top-of-atmosphere brightness temperatures, following standard GEE processing protocols. All LST layers were resampled to 10 m resolution to match the spatial reference grid of the snow classification outputs. Elevation-stratified mean, minimum, and maximum LST values were extracted for each of the four defined elevation zones to facilitate quantitative analysis of the relationship between thermal conditions and snow cover dynamics.

3.2.7 Snow Cover Change Detection

Inter-annual snow cover change was quantified using a post-classification comparison approach. Binary snow cover maps (Snow = 1, Non-Snow = 0) derived from the rule-based classifier were generated for each seasonal composite image for both the accumulation and depletion periods. All maps were standardized to the UTM projection, and classification scheme prior to differencing. Pixel-wise temporal difference maps were computed between consecutive annual pairs: 2019–2018, 2020–2019, 2021–2020, 2022–2021, 2023–2022, 2024–2023, and 2025–2024. Positive difference values indicate newly snow-covered pixels (snow expansion); negative values indicate pixels transitioning from snow to non-snow (snow retreat or depletion); and zero values represent unchanged pixels. The analysis was conducted separately for the accumulation and depletion seasons to capture season-specific patterns of inter-annual variability. For each annual pair, total snow-covered area (km^2), non-snow area, net snow cover change (km^2 and percentage), and the spatial distribution of gain, loss, and persistent snow were computed and mapped.

3.2.8 Elevation-Based Snow Dynamics Analysis

To investigate altitudinal controls on snow cover distribution, the study area was stratified into four elevation zones for profiling to study elevation based changes in snow cover by using the ALOS PALSAR DEM: Zone I (< 1500 m a.s.l.), Zone II (1500–2000 m a.s.l.), Zone III (2000–3000 m a.s.l.), and Zone IV (> 3000 m a.s.l.). For each classified seasonal snow map, the areal extent (km^2) of each snow class (Non-Snow, Dry Snow, and Wet Snow) was computed within each elevation zone. This yielded elevation-stratified time series spanning 2018–2025 for both seasons, enabling examination of how snow accumulation and melt behavior vary across the altitudinal gradient. Slope and aspect

layers derived from the DEM were additionally used to analyze the influence of terrain orientation on snow persistence, particularly the contrast between north-facing and south-facing slopes.

3.2.9 Climatic Correlation Analysis

To assess the influence of climate variability on observed snow cover dynamics, SPARTACUS gridded daily temperature and precipitation data were spatially aggregated over the study area for each seasonal window. Mean seasonal temperature (°C) and total seasonal precipitation (mm) were computed for each year and season. Pearson's correlation coefficient (r) was computed between these climatic variables and the fractional extent of each snow class (Dry Snow, Wet Snow) within each elevation zone. Statistical significance of correlations was assessed using two-tailed t-tests at the 0.05 significance level.

The SPARTACUS climate dataset covered seven meteorological observatory sub-regions within and adjacent to Hohe Tauern National Park: Central East Möll Valley, Central West Defereggental, East Ankogel Group, North Salzburg Rauris, Northeast Gastein Valley, Southeast Carinthia Lieser, and West Grossvenediger. The spatial distribution of temperature and precipitation across these sub-regions was analyzed to capture intra-park climatic heterogeneity.

To further characterize the relationship between snow class intensity and concurrent climatic conditions, a categorical correlation analysis was performed. Snow cover was partitioned into four intensity classes (Class 1 through Class 4) based on the magnitude of NDSI within the snow-covered domain, representing a gradient from low-intensity (Class 1: $0.4 < \text{NDSI} \leq 0.55$) to high-intensity snow cover (Class 4: $\text{NDSI} > 0.85$). Dry Snow and Wet Snow were analyzed separately. For each class, Pearson's r between precipitation and temperature was computed, and a linear regression model was fitted to quantify the directional relationship and its slope. This approach enabled examination of how the coupling between temperature and precipitation varied across snow intensity gradients during both accumulation and depletion phases.

3.2.10 Accuracy Assessment: AUC–ROC Analysis

The accuracy of the NDSI-based snow cover detection was evaluated using an Area Under the Curve–Receiver Operating Characteristic (AUC–ROC) analysis. This assessment compared the snow mapping performance of NDSI products derived from three satellite sensors operating at different spatial resolutions: Sentinel-2 MSI (10 m), Landsat 8/9 OLI (30 m), and MODIS (500 m). Sentinel-2 NDSI served as the primary classification platform, while Landsat and MODIS NDSI products were evaluated against it as comparative references.

Binary reference samples were generated using a stratified random sampling strategy applied to high-resolution optical imagery and supplemented by manual photo-interpretation. Stratified random points were collected, evenly distributed across snow and non-snow classes and stratified across elevation zones to ensure spatial representativeness. Ground truth labels were assigned as Snow (1) or Non-Snow (0) based on the majority class within a 3×3 pixel neighborhood to mitigate sub-pixel boundary effects.

ROC curves were generated by taking into account the Sentinel-2 ad reference line for each sensor by computing the True Positive Rate (TPR = sensitivity) and False Positive Rate (FPR = $1 - \text{specificity}$) across a range of NDSI threshold values from 0.10 to 0.90 at increments of 0.05. The AUC was computed as the integral of the ROC curve using the trapezoidal rule, with values approaching 1.0 indicating near-perfect discriminatory

capability. Sensor performance was compared on the basis of AUC magnitude, and the optimal NDSI threshold for each sensor was identified as the point on the ROC curve maximizing the Youden index (TPR – FPR).

3.2.11 Software and Processing Environment

All data acquisition, preprocessing, spectral index computation, classification, and spatial analysis were performed within **Google Earth Engine (GEE)**, a cloud-based geospatial analysis platform providing direct access to the full Copernicus and Landsat archives. Statistical post-processing, correlation analysis, and accuracy assessment were carried out in **Python** (version 3.13), using the libraries NumPy, pandas, matplotlib, and scikit-learn.

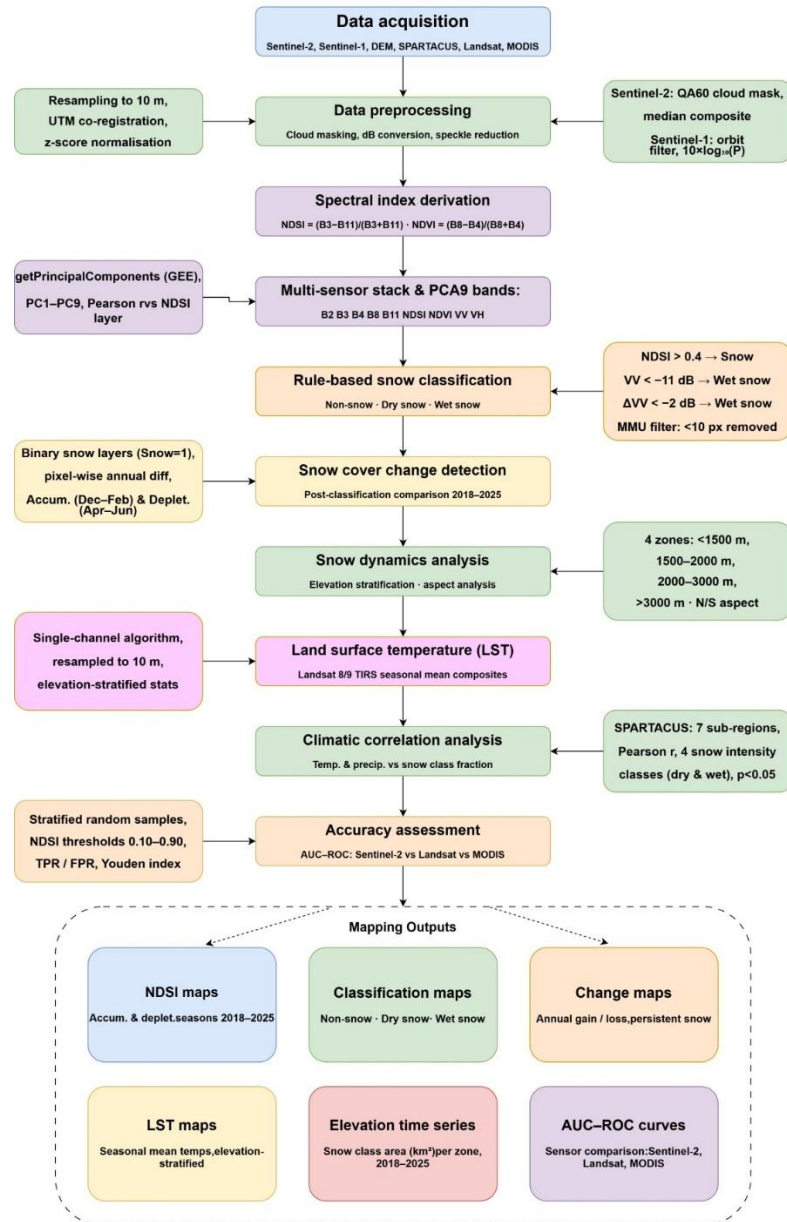


Figure 2 Methodological workflow for multi-sensor snow-cover mapping, snow-state classification, change detection, climatic analysis, and accuracy assessment

4 SNOW COVER DETECTION AND CLASSIFICATION

This chapter presents the results of the snow-cover detection and classification workflow described in Chapter 3. The analysis first evaluates the quality and consistency of the preprocessed Sentinel-2 and Sentinel-1 datasets and their co-registration prior to further analysis. Snow-cover extent is then assessed using NDSI-derived products for the accumulation and depletion seasons, followed by the classification of Non-Snow, Dry Snow, and Wet Snow conditions using the integrated optical and SAR information. The chapter further examines interannual changes in snow-cover distribution between 2018 and 2025 and evaluates the performance of NDSI-based snow detection using AUC-ROC analysis for Sentinel-2, Landsat 8/9, and MODIS. Together, these results provide the basis for assessing the spatial and temporal variability of snow cover across the elevation zones of Hohe Tauern National Park.

4.1 Preprocessing Verification

4.1.1 Sentinel-2 Preprocessing Verification

Prior to analysis, the quality of preprocessed input datasets was verified to ensure spatial consistency, radiometric integrity, and freedom from significant cloud or speckle contamination. Sentinel-2 QA60-based cloud masking effectively removed the majority of cloud and cloud shadow pixels from the composites. Median temporal compositing over the defined seasonal windows further suppressed residual noise and minimized the influence of transient atmospheric effects, yielding stable surface reflectance representations for each season and year. Visual inspection confirmed that cloud and shadow contamination were reduced to negligible levels in all 16 Sentinel-2 seasonal composites.

4.1.2 Sentinel-1 Preprocessing Verification

Sentinel-1 SAR preprocessing produced VV and VH backscatter composites free from systematic geometric distortions. Temporal median compositing over each seasonal window effectively suppressed speckle noise, as evidenced by spatially coherent backscatter distributions across the AOI. Backscatter values in the accumulation season composites ranged from approximately -15 to -5 dB in VV polarization, consistent with expectations for snow-covered alpine terrain at C-band frequencies.

4.1.3 Multi-Sensor Co-registration and Normalization

Co-registration between Sentinel-1 and Sentinel-2 composites was achieved at sub-pixel accuracy through GEE's common reprojection framework, and z-score normalization confirmed that all nine input bands contributed approximately equally to the PCA prior to transformation, with standardized means of 0 and unit variance across all channels.

4.2 NDSI-Based Snow-Cover Detection Results

4.2.1 Accumulation Period NDSI:

The seasonal mean NDSI maps for the accumulation period (Figure 3) reveal a clear and consistent altitudinal gradient in snow cover intensity across the Hohe Tauern National Park over the 2018–2025 study period. High-elevation areas (> 2000 m a.s.l.)

consistently exhibit NDSI values approaching the upper bound of the index (≥ 0.90), indicating persistent and optically dense snow cover throughout all years. Lower-elevation zones (< 1500 m) display markedly lower NDSI values, reflecting partial or ephemeral snow cover subject to recurrent melt events.

Over the early years of the study period (2018–2022), accumulation-season NDSI values at mid-to-high elevations increased progressively, reaching peak values of approximately 0.99 by 2022, reflecting enhanced snow persistence and accumulation. This trend suggests a period of relatively favorable accumulation conditions at higher altitudes during this interval. In contrast, the later years (2024–2025) exhibit a measurable shift in snow dynamics: NDSI values in valley and foothill zones decline substantially, with minimum values reaching approximately -0.55 to -0.66 in the lowest elevation zones. These strongly negative NDSI values in low-altitude areas indicate the predominance of vegetated or bare soil surfaces with negligible snow cover, consistent with earlier melt onset or reduced snowfall accumulation at lower elevations. High-elevation NDSI values remain elevated (> 0.90) throughout the full study period, indicating that the upper alpine zone retains its capacity for sustained winter snowpack even as lower elevations experience declining snow persistence. Collectively, the accumulation-season NDSI time series indicates a transition toward elevation-dependent snow retention, with an accelerating loss of snow cover at lower altitudes and relative resilience at high elevations.

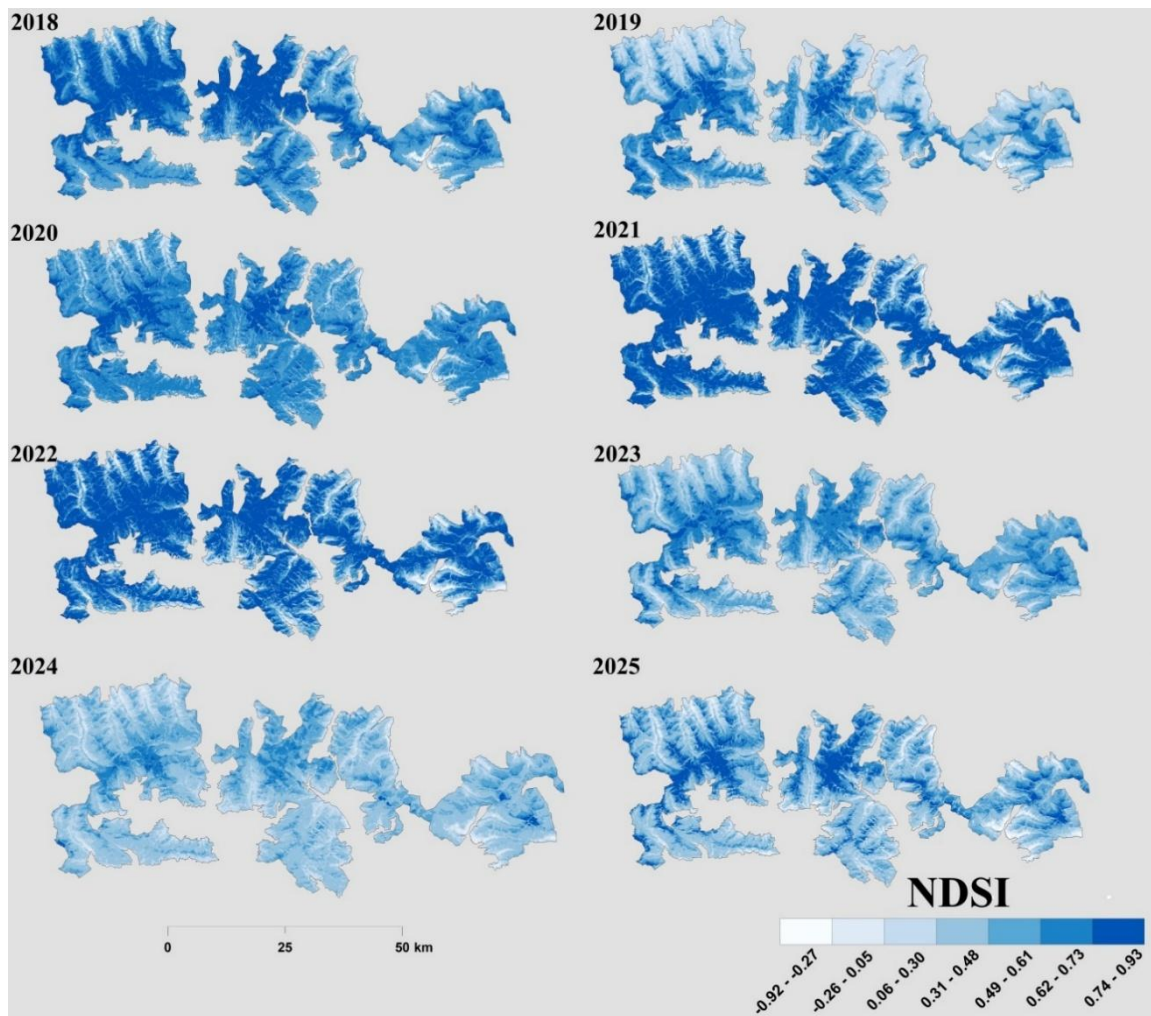


Figure 3 Accumulation-season Normalized Difference Snow Index maps in the study area, 2018–2025.

4.2.2 Depletion Period NDSI:

Depletion-season NDSI maps (Figure 4) show a comparable altitudinal structure, with high-elevation zones maintaining NDSI values between 0.90 and 0.96 throughout the study period, reflecting persistent residual snowpack in spring, while valley floors display NDSI values indicative of snow-free or largely snow-depleted conditions. Note that negative NDSI values in these composites represent non-snow surfaces—predominantly vegetation or bare soil—and do not carry quantitative snow cover meaning; their spatial extent serves as an indicator of the snow-free fraction within each seasonal composite image.

During the early phase (2018–2020), depletion-season NDSI values at high elevations ranged from 0.93 to 0.96, consistent with substantial residual snowpack after the main accumulation season. Snow depletion in valley areas was already pronounced in all years, with low or negative NDSI values persisting across lower-elevation areas throughout the depletion window, indicating that snowmelt is a consistent early-season process at these altitudes regardless of inter-annual accumulation variability. A modest reduction in peak high-elevation NDSI is observed during 2021–2022, where maximum values decline to approximately 0.91, suggesting slightly reduced net accumulation entering the depletion period in these years. A partial recovery is evident in 2023, with high-elevation NDSI values returning to approximately 0.95; however, low-elevation snow cover does not recover comparably. The 2024–2025 period shows stabilization, with high-elevation NDSI values near 0.93 and continued low values in valley areas. These patterns indicate that while the upper alpine zone sustains relatively consistent depletion-season snowpack, mid- and lower-elevation zones experience progressive loss of spring snow cover, consistent with a warming-driven upward shift of the seasonal snowline.

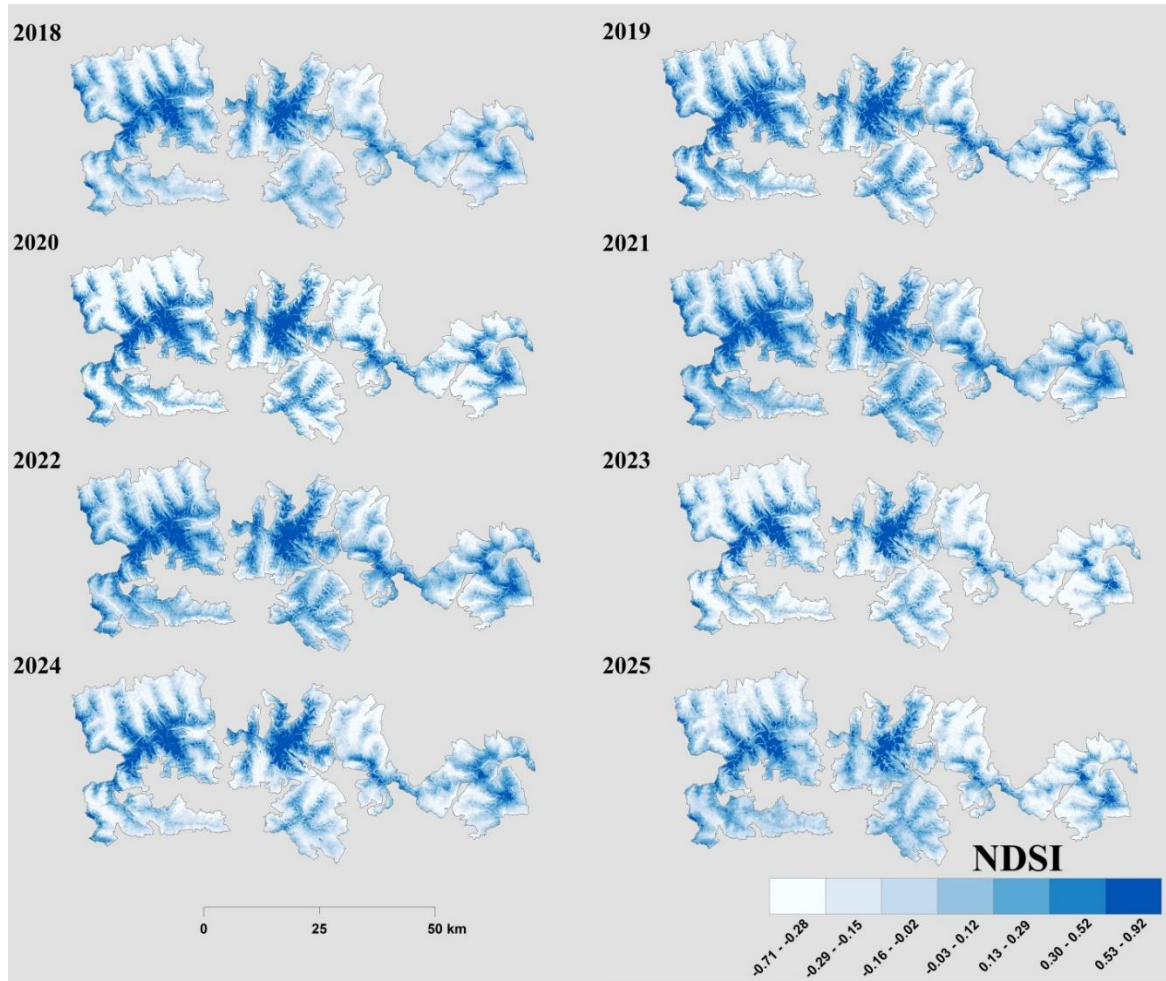


Figure 4 Depletion-season Normalized Difference Snow Index maps of the study area, 2018–2025.

4.3 Snow Cover Classification Results

4.3.1 Accumulation-Season Classification

The rule-based classification of accumulation-season composites (Figure 5) reveals pronounced temporal and elevation-dependent variability in the distribution of Non-Snow, Dry Snow, and Wet Snow across the Hohe Tauern National Park over 2018–2025. Overall, the study period is characterized by a progressive reduction in Dry Snow extent and an increasing prevalence of Wet Snow, particularly at mid-elevations, alongside a gradual expansion of Non-Snow area at lower altitudes.

In the lowest elevation zone (Zone I: < 1500 m), Non-Snow and Wet Snow are the dominant classes throughout the study period. Non-Snow area increases progressively over time, peaking at approximately 42.49 km² in 2023. Dry Snow coverage is consistently minimal in this zone, confined to occasional cold-spell events. In Zone II (1500–2000 m), notable inter-annual variability is observed, with Non-Snow area reaching its maximum extent of approximately 153.01 km² in 2023, coinciding with reduced Wet Snow coverage. A dry Snow exhibit localized increases after 2020 but does not offset the broader trend of

declining snow persistence in this zone, which may reflect increasing precipitation phase transitions from snowfall to rainfall under warmer accumulation-season temperatures.

In Zone III (2000–3000 m), snow cover is more persistent, with Wet Snow consistently dominating despite a slight declining trend across the study period. Non-Snow area remains minimal, confirming sustained snow presence at this elevation, though Dry Snow exhibits inter-annual variability and a declining tendency from 2022 onward. At the highest elevations (Zone IV: > 3000 m), snow cover is largely stable. However, Dry Snow area shows a substantial reduction after 2020 while Wet Snow remains relatively constant, with values transitioning from approximately 80.48 km² in 2020 to 79.37 km² in 2025. This shift from Dry to Wet Snow at the highest elevations is particularly noteworthy, as it indicates that warming is penetrating even the most elevated portions of the park, altering snow crystal structure and liquid water content even where total snow area remains stable.

Collectively, the accumulation-season classification reveals a transition toward reduced total snow extent, a downslope shift of the Dry Snow domain, and increasing Wet Snow prevalence across all elevation zones, suggesting a systematically warming accumulation season with implications for snowpack water equivalent and downstream hydrology.

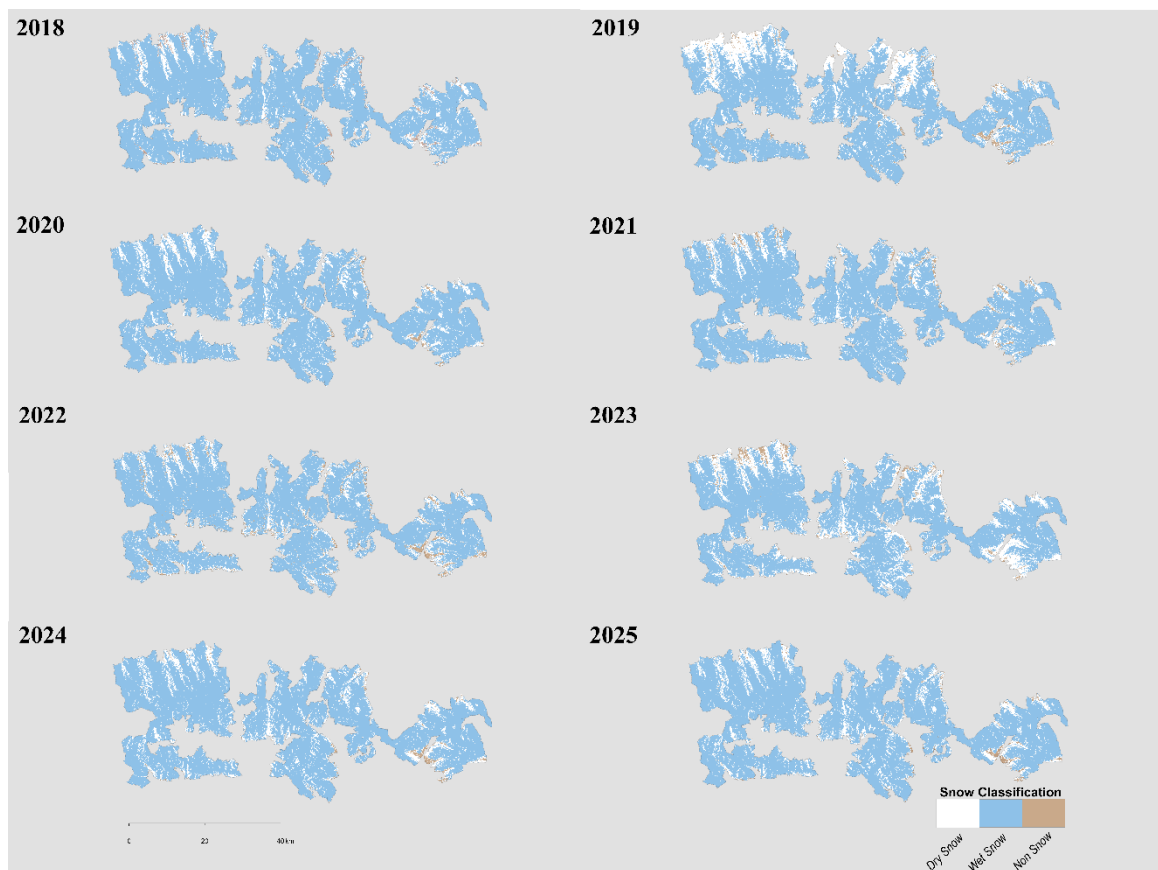


Figure 5 Spatial distribution of Non-Snow, Dry Snow, and Wet Snow classes during the accumulation season in Hohe Tauern National Park, 2018–2025.

4.3.2 Depletion-Season Classification

The depletion-season classification results (Figure 6) document a progressive transition in snow regime over 2018–2025, with distinct elevation-dependent and aspect-dependent patterns that differentiate the depletion season from the accumulation season. Non-Snow, Dry Snow, and Wet Snow exhibit a clear temporal succession consistent with an advancing melt season.

In the early phase (2018–2019), Dry Snow was largely confined to elevations above 2500 m a.s.l., while Non-Snow dominated valley floors and lower slopes. During 2020–2021, Wet Snow expanded substantially in mid-elevation zones (Zone II–III), reflecting increased liquid water content within the snowpack as temperatures approached the melting point. By the mid-to-late phase (2022–2025), Non-Snow areas had expanded significantly across Zones I and II, while Dry Snow retreated to isolated high-elevation patches (> 3000 m). Wet Snow remained the dominant snow class at mid-to-high elevations throughout the depletion season, underlining its role as the transitional state between frozen snowpack and complete ablation.

A spatially consistent aspect dependency was identified in the classification outputs: north-facing slopes (aspect 315°–360° and 0°–45°) retained proportionally greater Dry Snow coverage compared to south-facing slopes (aspect 135°–225°), which exhibited substantially larger Non-Snow extents. This asymmetry reflects differential solar radiation loading and its role in accelerating surface melt on sun-exposed aspects during the depletion window. In Zone III (2000–3000 m), Dry Snow declined sharply in 2024 to approximately 1.36 km², representing its lowest value within the depletion season record. This result, combined with the concurrent persistence of Wet Snow, indicates that the transition from dry to wet snowpack is occurring earlier in the melt season and at progressively higher elevations in recent years.

The depletion-season results provide complementary evidence to the accumulation-season findings, with both seasons indicating reduced Dry Snow extent, earlier transitions to Wet Snow, and increasing snow loss at lower elevations. These patterns are consistent with warming-related changes in the Hohe Tauern alpine snow regime.

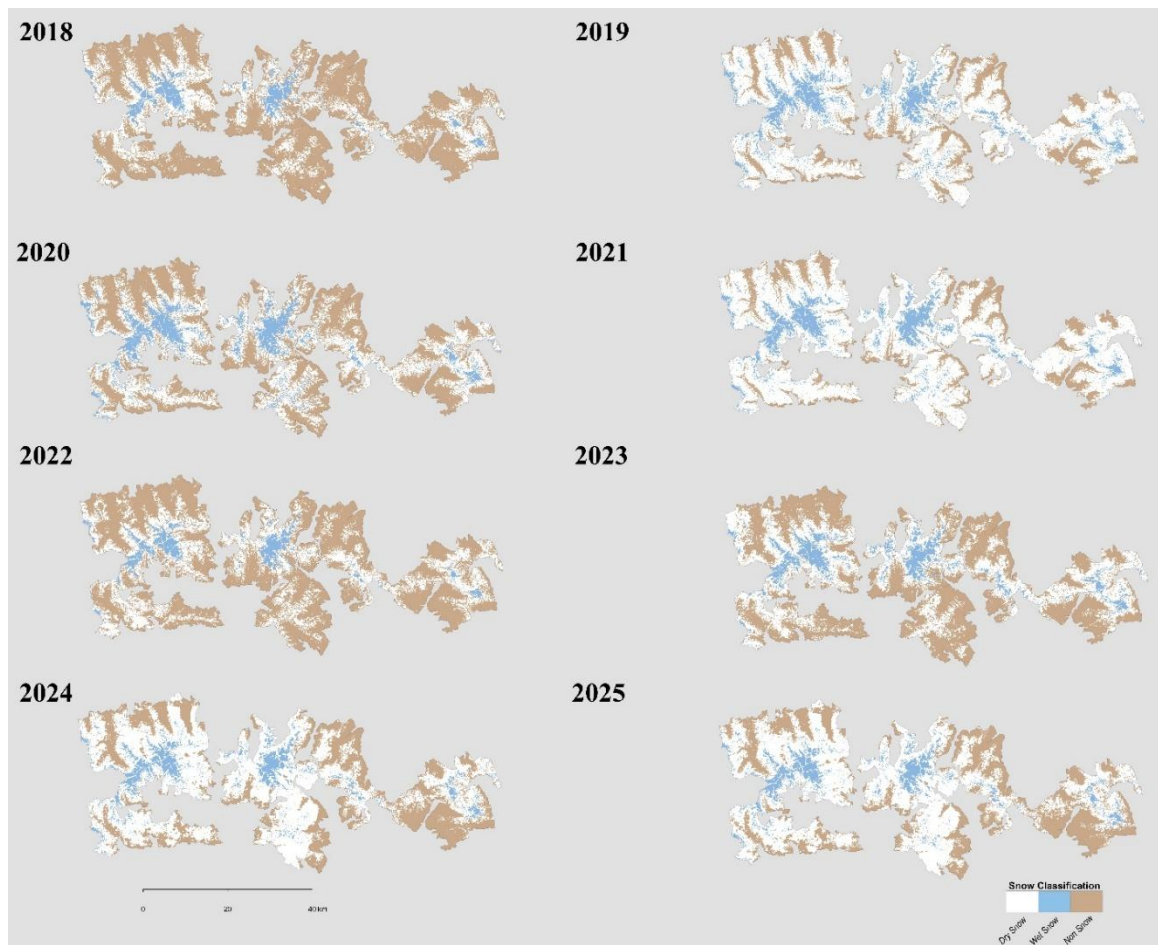


Figure 6 Spatial distribution of Non-Snow, Dry Snow, and Wet Snow classes during the depletion season in the study area, 2018–2025.

4.4 Interannual Snow-Cover Change

4.4.1 Accumulation-Season Snow-Cover Change, 2018–2025

Inter-annual snow cover change maps (Figures 7 and 8) quantify the spatial distribution and magnitude of snow gain, snow loss, and persistent snow across seven consecutive annual pairs for both the accumulation and depletion seasons. During the accumulation season, net snow cover change was predominantly negative (indicating snow loss) at elevations below 2000 m across most annual transitions, with the largest losses occurring between 2022–2023 in Zones I and II. Snow expansion events were detected primarily above 2500 m in the 2020–2021 and 2021–2022 pairs, consistent with the elevated NDSI values recorded during those accumulation seasons.

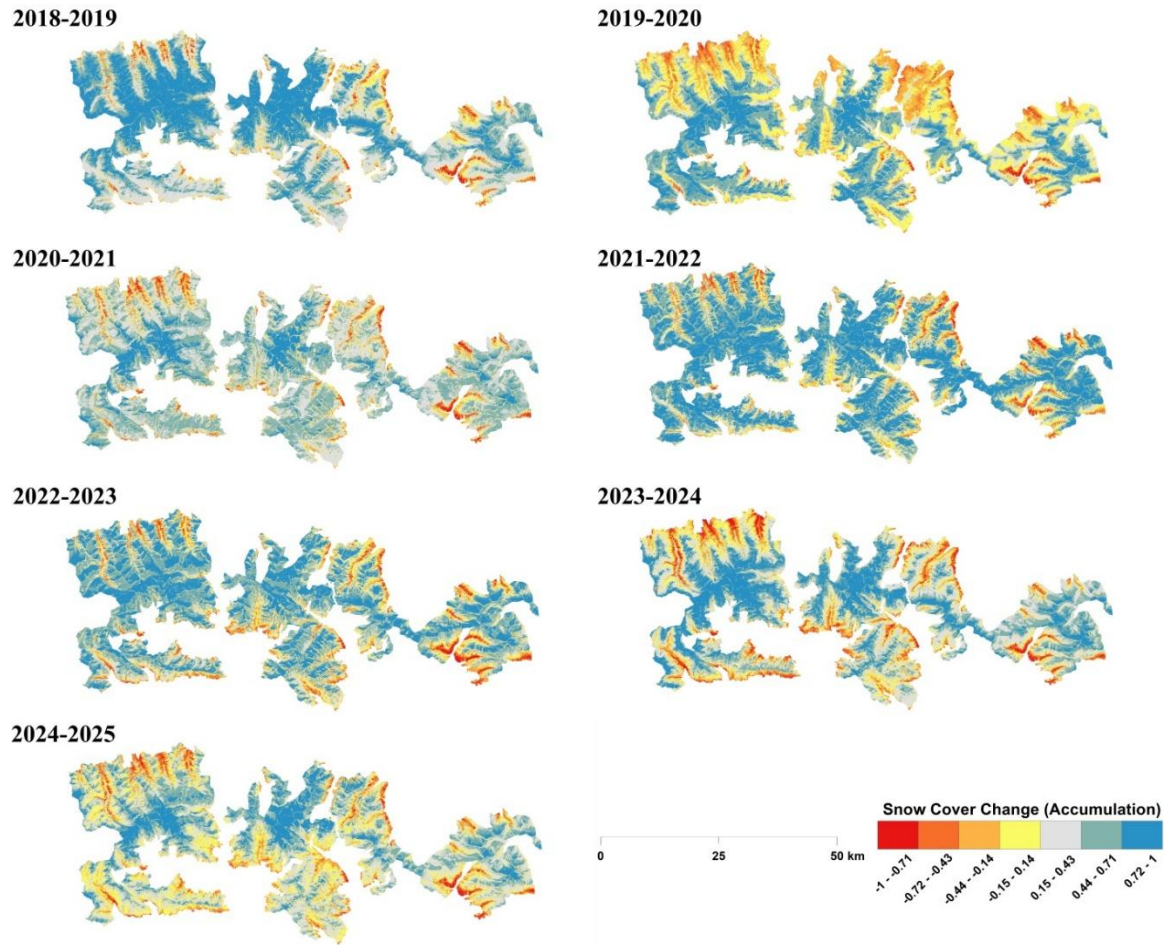


Figure 7 Interannual changes in accumulation-season snow cover between consecutive years in the study area, 2018–2025. Negative values indicate decreases, while positive values indicate increases in snow cover.

4.4.2 Depletion-Season Snow-Cover Change, 2018–2025

During the depletion season, the change maps reveal a consistent pattern of snow area reduction at lower elevations across all annual pairs, with the rate of retreat intensifying in the latter half of the study period (2022–2025). Persistent snow areas—pixels classified as snow-covered in both the earlier and later year of each pair—were concentrated above 2500 m in all pairs, confirming the stability of high-elevation snowpack even as lower elevations experienced net loss. The spatial distribution of change clearly delineates an upward shift in the consistent snow limit across the study period, corroborating the NDSI and classification analyses.

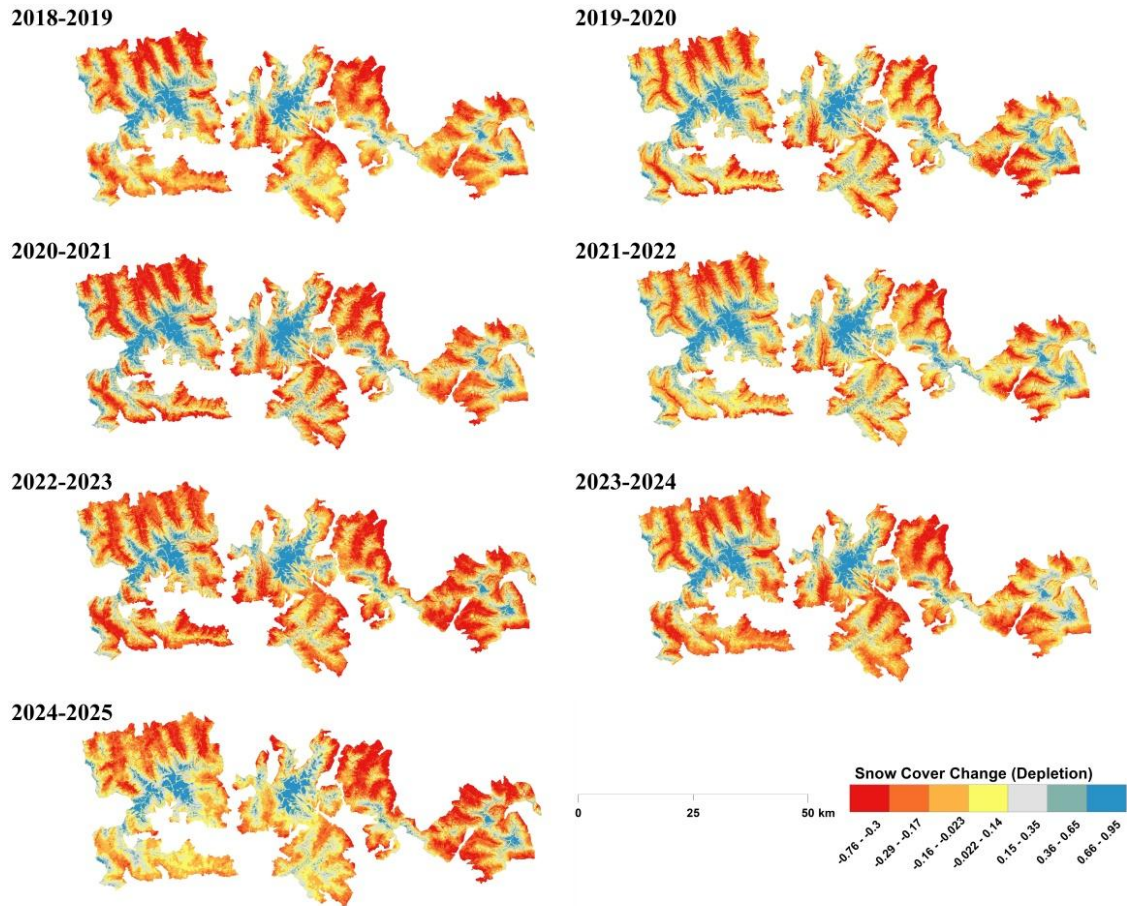


Figure 8 Interannual changes in depletion-season snow cover between consecutive years the study area, 2018–2025. Negative values indicate decreases, while positive values indicate increases in snow cover.

4.5 Accuracy Assessment

4.5.1 AUC–ROC Comparison of Sentinel-2, Landsat 8/9, and MODIS

The AUC–ROC analysis (Figure 9) evaluated the discriminatory capability of NDSI products derived from Sentinel-2 (10 m), Landsat 8/9 OLI (30 m), and MODIS (500 m) in distinguishing snow-covered from snow-free pixels. The results revealed notable differences in threshold-independent classification performance among the three sensors. Sentinel-2 achieved the highest AUC value of 0.98, indicating excellent discrimination ability and a robust capacity to separate snow-covered surfaces from non-snow backgrounds across the full range of NDSI thresholds. This superior discriminatory performance can be attributed to its fine spatial resolution and enhanced spectral characteristics, which enable more precise differentiation of snow-covered surfaces in complex alpine environments.

Among the remaining sensors, Landsat-based NDSI achieved an AUC value of 0.8321, demonstrating strong discriminatory capability and confirming the suitability of its 30 m spatial resolution for snow-cover mapping in mountainous terrain. The MODIS NDSI product yielded an AUC value of 0.7028, indicating acceptable discriminatory power but with comparatively weaker separation of snow and non-snow classes. The reduced

discriminatory performance of MODIS is primarily associated with its coarser 500 m spatial resolution, which increases mixed-pixel effects and sub-pixel heterogeneity in topographically complex landscapes.

The ROC curves further demonstrate that Sentinel-2 consistently achieved the most favorable balance between sensitivity (true positive rate) and specificity (1 – false positive rate), followed by Landsat and MODIS. The superior discrimination of Sentinel-2 reflects its ability to capture fine-scale snow distribution patterns, while Landsat provides a reliable compromise between spatial detail and temporal coverage. In contrast, MODIS with 500 m of spatial resolution remains effective for broad-scale snow monitoring despite its weaker discriminatory performance.

Overall, all three sensors demonstrated operationally useful discriminatory capability for snow-cover mapping, with AUC values exceeding 0.70. Sentinel-2 exhibited the highest discrimination ability, making it particularly suitable for detailed snow-cover assessments in heterogeneous alpine terrain such as Hohe Tauern National Park. Landsat offers strong discriminatory performance for regional-scale applications, whereas MODIS retains advantages in near-daily temporal coverage and global availability, making it valuable for large-scale and time-sensitive snow-monitoring programs.

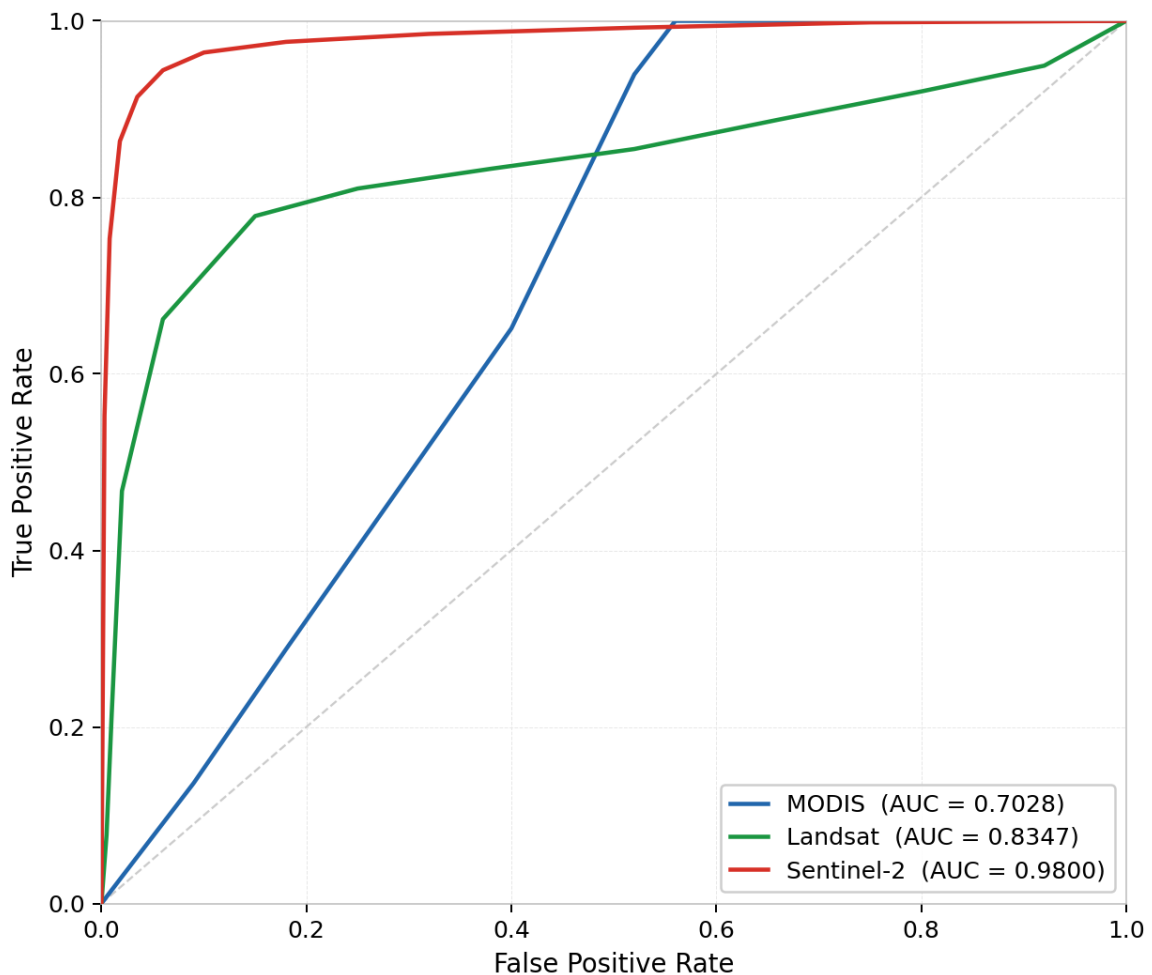


Figure 9 Receiver operating characteristic curves comparing the performance of Sentinel-2, Landsat 8/9, and MODIS for NDSI-based discrimination between snow and non-snow surfaces.

5 SNOW COVER DYNAMICS AND CLIMATIC DRIVERS

This chapter examines the spatial and temporal dynamics of snow cover in relation to climatic and topographic variability across Hohe Tauern National Park during the 2018–2025 study period. The analysis first investigates seasonal land surface temperature patterns across the defined elevation zones and then evaluates interannual variability in temperature and precipitation. Relationships between climatic variables and the spatial extent of Dry Snow and Wet Snow are subsequently assessed for individual elevation classes to identify differences in the relative influence of temperature and precipitation across the study area. These analyses provide a climatic context for the snow-cover patterns identified in Chapter 4 and support the interpretation of elevation-dependent changes in snow persistence, snow state, and snow loss.

5.1 Land Surface Temperature

5.1.1 Accumulation-Season Land Surface Temperature

Seasonal mean LST maps for the accumulation period (Figure 10) reveal a progressive warming trend across all elevation zones in Hohe Tauern National Park. At the region-wide scale, maximum accumulation-season LST increased from approximately 6.11°C in 2018 to 7.57°C in 2025, while minimum temperatures also became less extreme over the same period. At Zone I (< 1500 m), mean LST increased from -2.09°C in 2018 to approximately +0.39°C in 2025, representing a warming of ~2.5°C and crossing the freezing threshold — a critical change in the surface thermal regime that directly reduces snow persistence. Zone II (1500–2000 m) exhibited a mean LST increase from -3.82°C in 2018 to -1.30°C in 2025, while Zone III (2000–3000 m) and Zone IV (> 3000 m) showed increases from -6.62°C to -4.26°C and from -10.07°C to -7.89°C, respectively, over the same period.

A temporary reduction in peak LST was recorded in 2021 (~3.47°C), representing a short-term departure from the overall warming trajectory. This anomaly coincides with a slight recovery in snow extent and elevated NDSI values observed in the same year, suggesting a direct thermal control on seasonal snow conditions. The broader warming signal at all elevations aligns closely with the spatial pattern of declining Dry Snow extent and expanding Non-Snow area documented in the classification analysis, reinforcing a physically consistent interpretation of warming-driven snow cover change.

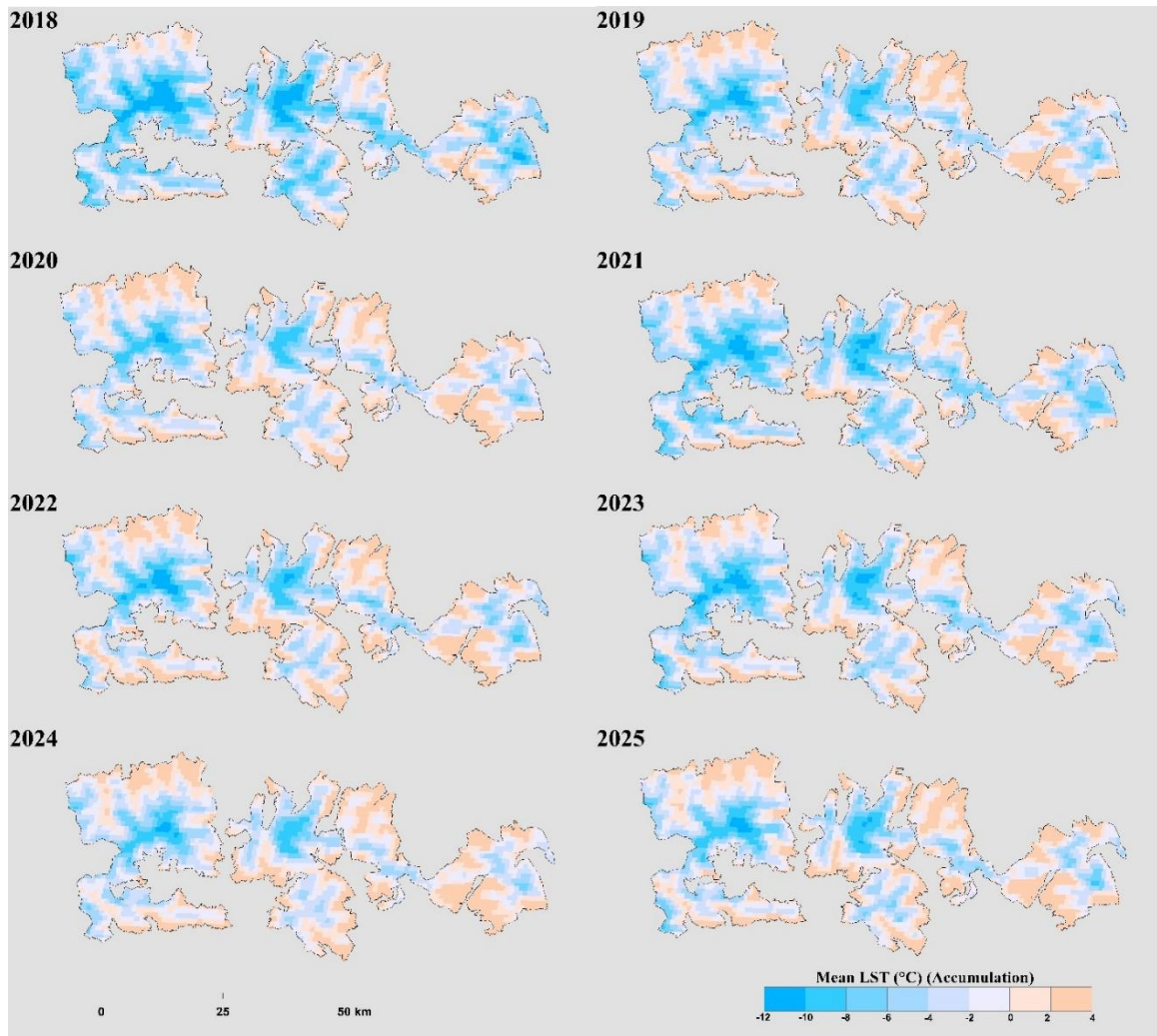


Figure 10 Mean accumulation-season land surface temperature maps of the study area for 2018–2025.

5.1.2 Depletion-Season Land Surface Temperature

Depletion-season LST maps (Figure 11) confirm a consistent warming pattern, albeit superimposed on stronger inter-annual variability than in the accumulation season. Maximum LST values exceeded 21°C in all years, with a gradual upward trend reaching above 22°C by 2024–2025. Minimum temperatures, which control overnight refreezing of the snowpack, increased progressively across all zones. In Zone I (< 1500 m), minimum temperatures stabilized above 0°C during the depletion window in the latter half of the study period (2022–2025), substantially reducing the nighttime refreezing capacity of the snowpack and accelerating net melt. A brief cooling anomaly in 2021 (minimum temperatures slightly below 0°C) is consistent with the temporary NDSI recovery and Dry Snow persistence observed in that year.

At the highest elevations (Zone IV, > 3000 m), mean LST values remained well below freezing throughout the depletion season, confirming the thermal stability of high-elevation snowpack. Nevertheless, a subtle but consistent upward shift in Zone IV mean temperatures (approximately 2–3°C, over the 2018–2025 study period) suggests that even the highest portions of the park are not immune to the regional warming trend, with potential long-term consequences for perennial snowfields and glacial margins.

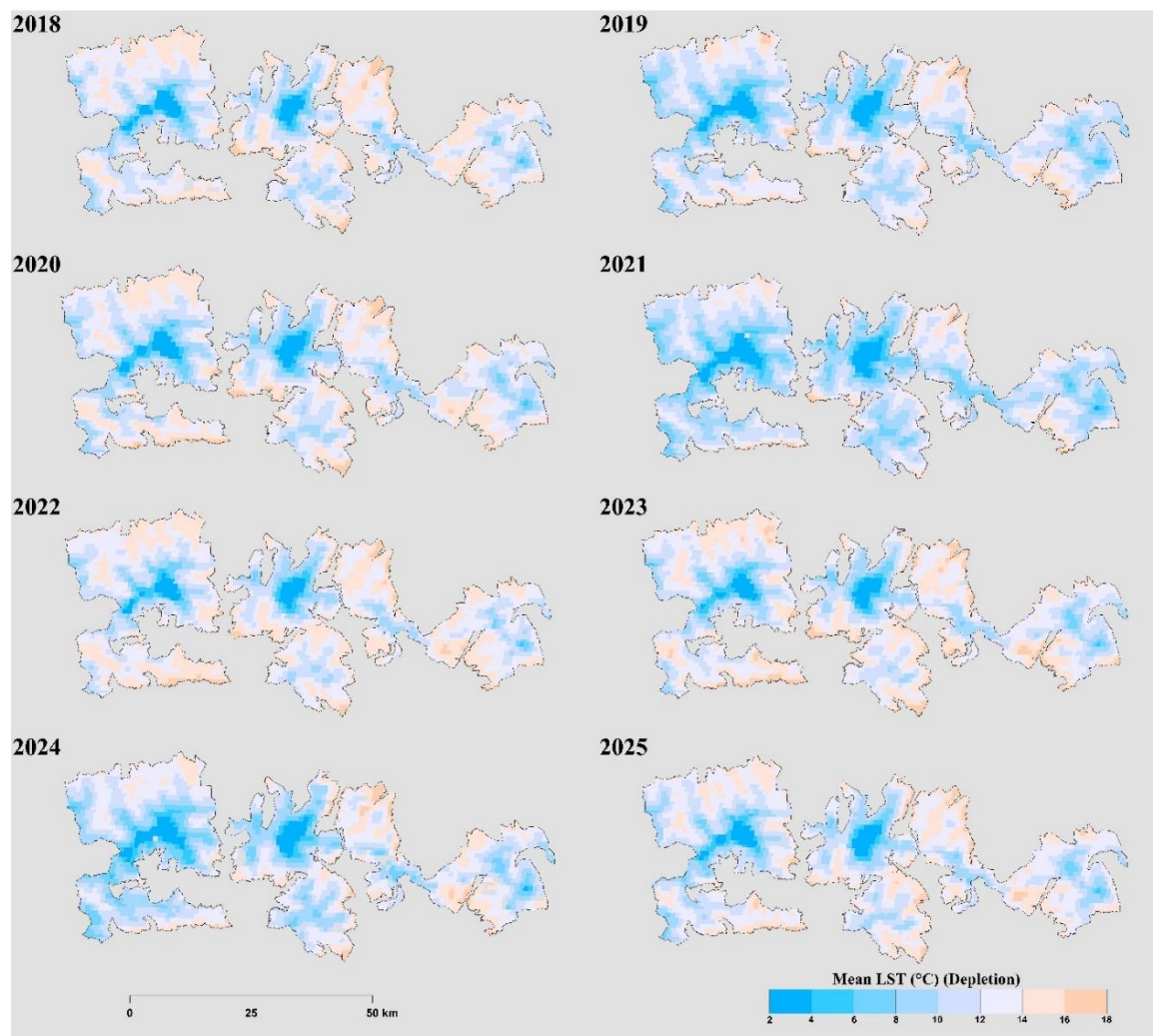


Figure 11 Mean depletion-season land surface temperature maps of the study area for 2018–2025.

5.2 Temperature and Precipitation Variability

The SPARTACUS temperature record for the Hohe Tauern National Park region (Figure 12) demonstrates a strong and well-defined seasonal cycle characteristic of high-altitude alpine environments, with mean annual temperatures close to 0°C and a wide annual range extending from strongly negative winter minima to mild summer maxima. Winter months (December–February) are consistently characterized by sub-zero temperatures, establishing the thermal prerequisite for sustained snowpack accumulation. Spring (March–May) represents a transitional thermal phase, during which temperatures rise progressively toward the melting threshold, initiating snowmelt. Summer (June–August) temperatures remain mild in the high-alpine zone, consistent with the maintenance of snow and ice at the highest elevations while promoting progressive melt at lower altitudes. Autumn cooling reinstates sub-zero conditions by late October to November, reinitiating the accumulation cycle.

Precipitation across the park exhibits a bimodal seasonal distribution, with a primary maximum during summer months associated with convective activity, and a secondary maximum in autumn linked to frontal systems. Winter precipitation is comparatively

lower in frequency but occurs predominantly as snowfall given the prevailing sub-zero temperatures, contributing to snowpack development. A notable high-precipitation event in late 2019 contributed disproportionately to that year's annual total, highlighting the episodic nature of alpine precipitation and its potential influence on snowpack variability.

The correlation between seasonal temperature and precipitation was weak and positive ($r \approx 0.2\text{--}0.35$), reflecting the tendency for higher precipitation to occur during relatively warmer summer months when convective activity is greatest. This weak coupling indicates that precipitation phase rather than total amount is the critical determinant of snowpack development, with winter precipitation contributing to snow accumulation only when temperatures are sufficiently below the melting threshold. These results underscore that snow cover variability in Hohe Tauern National Park is primarily governed by the interaction of seasonal temperature regimes with precipitation timing, rather than by monotonic trends in either variable alone.

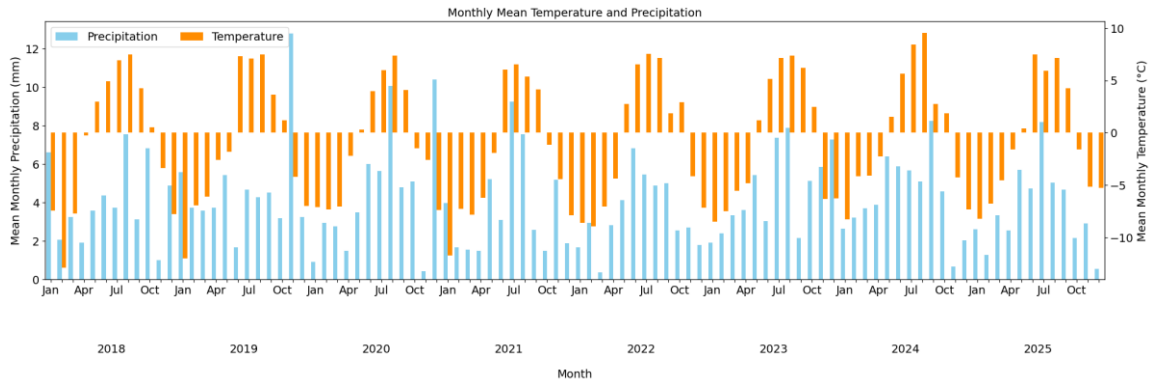


Figure 12 Monthly mean temperature and precipitation in the study area from 2018 to 2025.

5.3 Relationships Between Climate Variables and Snow Classes

5.3.1 Temperature versus Dry Snow Area by Elevation Zone

The negative relationship between temperature and dry snow cover observed at the basin scale exhibits considerable variability across elevation zones (Figure 13). In the lowest elevation band (<1500 m), the scatterplot shows substantial dispersion and a negligible coefficient of determination (R^2), indicating a weak association between temperature and dry snow extent. This pattern reflects the episodic nature of dry snow occurrence at lower elevations, where snow persistence is strongly influenced by short-term temperature fluctuations and transient weather conditions rather than seasonal temperature averages. The 1500–2000 m band shows a slightly stronger negative signal, consistent with this elevation range straddling the climatological rain–snow transition zone where inter-annual temperature variability has a more direct control on snow phase. In the 2000–3000 m and >3000 m bands, the dry snow signal remains consistently small in absolute area and shows little systematic temperature dependence, reinforcing the interpretation that these high-altitude zones maintain cold enough conditions throughout the accumulation season that mean temperature variability within the observed range does not meaningfully perturb dry snow extent. The near-flat regression lines at high elevations confirm that temperature-driven changes in dry snow are most likely to manifest first at the lower elevation margins of the snowpack.

Temperature vs. Dry Snow Area by Elevation Class (2018-2025)

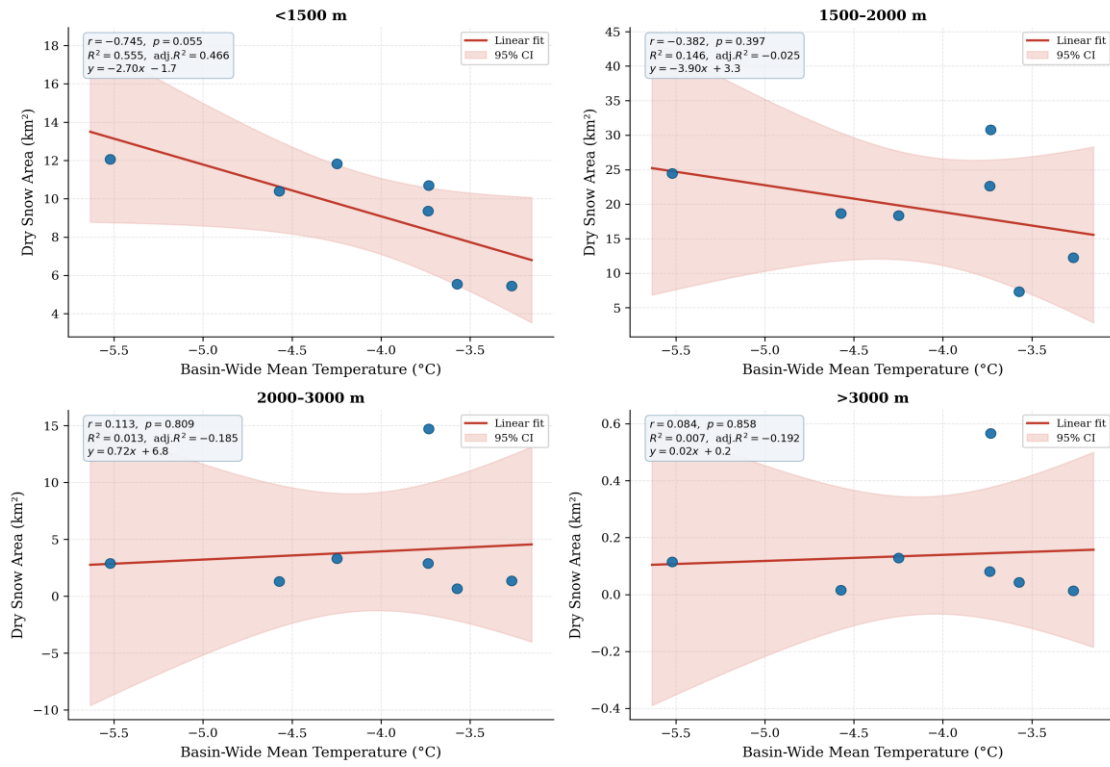


Figure 13 Four-panel scatter plots of basin-wide mean accumulation-season temperature (°C) versus dry snow area (km²) for each elevation class across the study area.

5.3.2 Temperature versus Wet Snow Area by Elevation Zone

The relationship between temperature and wet snow cover exhibits pronounced elevation-dependent variability (Figure 14). In the <1500 m and 1500–2000 m elevation bands, wet snow area shows moderate negative correlations with temperature, indicating that warmer conditions at lower elevations enhance snowmelt processes and increase the likelihood of rain-on-snow events. These processes accelerate the transition of wet snow into snow-free conditions, thereby reducing the extent and persistence of wet snow cover within these elevation zones. In contrast, the 2000–3000 m band — which dominates the total wet snow signal with values between approximately 1,150 and 1,270 km² shows a relatively flat regression with high scatter, suggesting that the wet snow area at this critical elevation band is insensitive to mean temperature variability within the observed range. This plateau-like behavior arises because temperatures at 2000–3000 m remain sufficiently below freezing throughout the accumulation season even in the warmest years of the study period. The >3000 m band similarly shows minimal temperature sensitivity. These results collectively suggest that a warming threshold exists somewhere in the 1500–2000 m range at which point temperature begins to meaningfully erode wet snow area, with implications for the sensitivity of Hohe Tauren snowpack to continued Alpine warming.

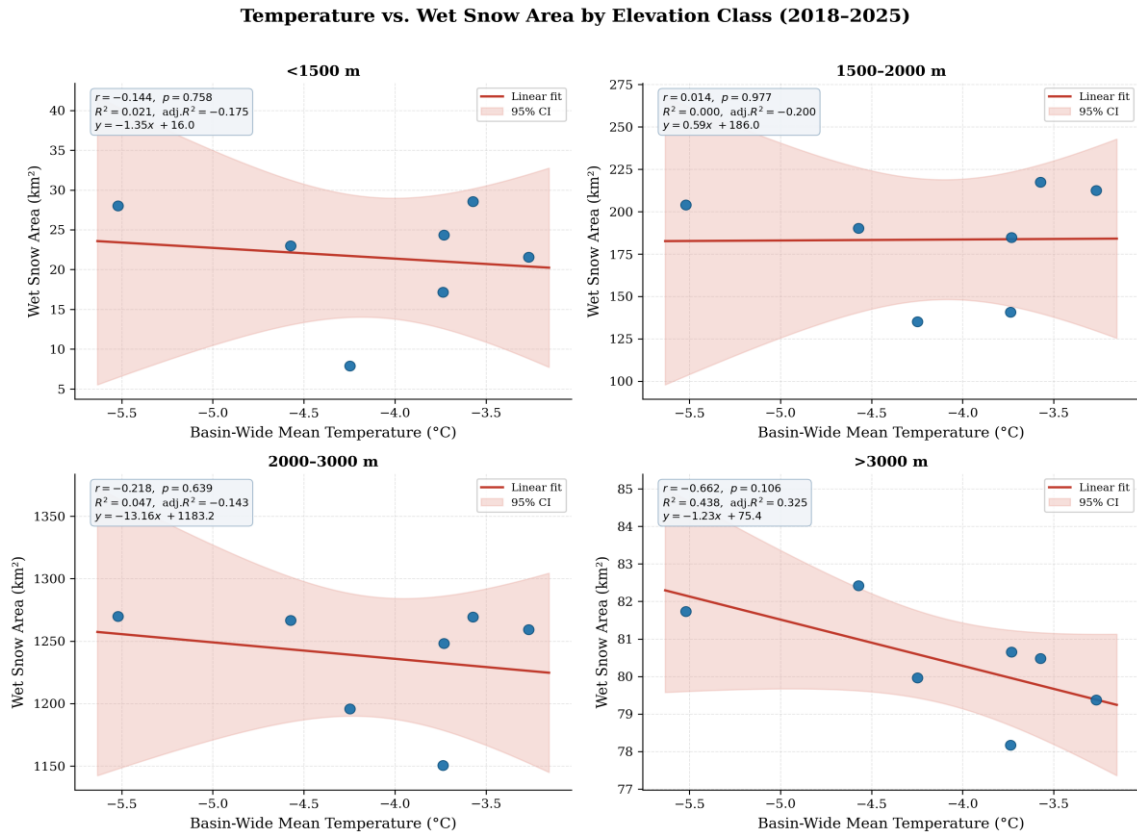


Figure 14 Four-panel scatter plots of basin-wide mean accumulation-season temperature (°C) versus wet snow area (km²) for each elevation class across the study area.

5.3.3 Precipitation versus Dry Snow Area by Elevation Zone

The elevation-dependent scatterplots (Figure 15) demonstrate that the weak basin-scale negative correlation between precipitation and dry snow cover is not uniform across altitudinal zones. In the <1500 m elevation band, data points are widely scattered with no evident trend, indicating that dry snow occurrence at lower elevations is highly episodic and governed by complex meteorological conditions. This substantial variability

highlights the influence of factors beyond precipitation, particularly temperature fluctuations and rapid snowmelt processes, in controlling dry snow distribution at low elevations. The 1500–2000 m band shows a slightly more negative slope, potentially reflecting that higher precipitation events at this elevation involve warmer synoptic conditions that reduce dry snow fractions. Most notably, the 2000–3000 m and >3000 m bands exhibit near-flat regression lines with very low R^2 values, confirming that precipitation amount has little impact on whether snow is classified as dry at these altitudes — the consistently cold temperatures there ensure all precipitation falls as snow and maintains dry snow characteristics regardless of precipitation intensity. The strong year-to-year variability in dry snow area at the 2000–3000 m band (ranging from 0.66 to 14.7 km²) likely reflects inter-annual differences in wind scouring, redistribution, and transient warm spells rather than total precipitation volume.

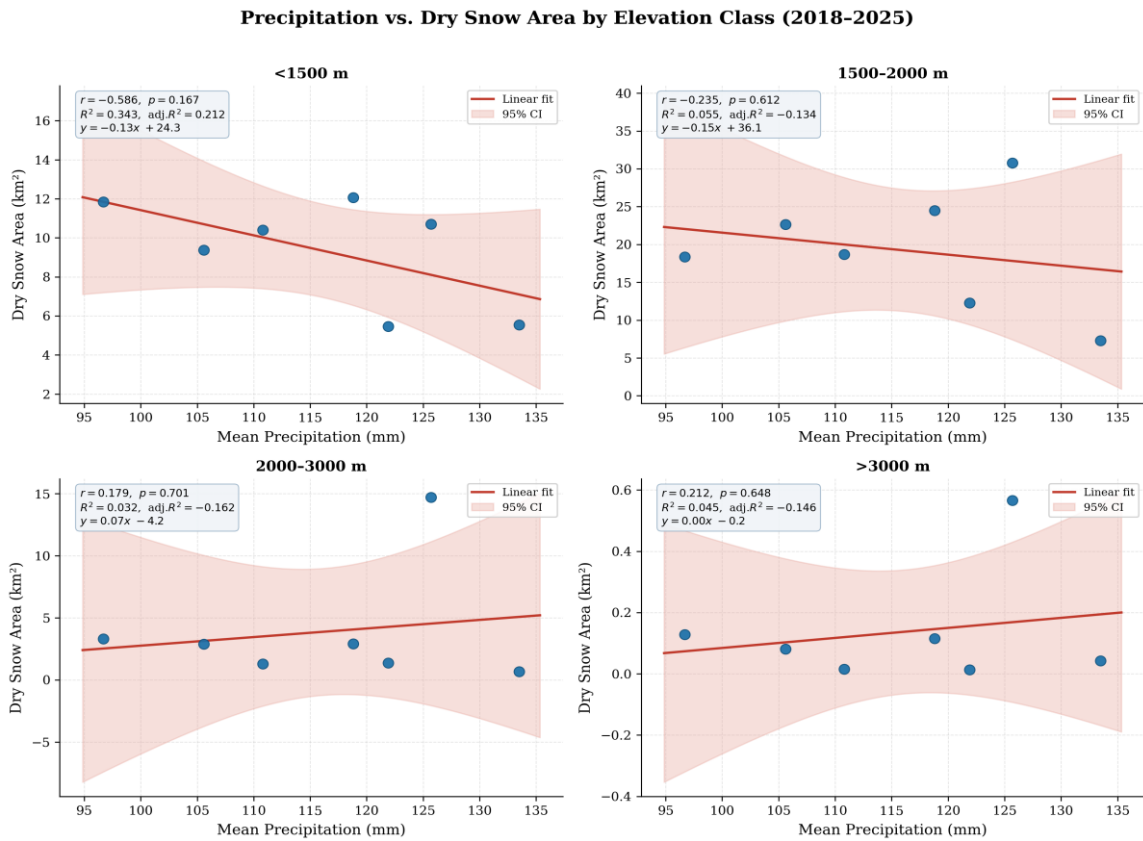


Figure 15 Four-panel scatter plots of mean accumulation-season precipitation (mm) versus dry snow area (km²) for each elevation class across the study area.

5.3.4 Precipitation versus Wet Snow Area by Elevation Zone

The comparison of the Precipitation with the wet-snow area shows the most scientifically compelling elevation-stratified results (Figure 16). The strong positive precipitation–wet snow relationship identified at the basin scale is most clearly expressed in the 2000–3000 m elevation band, which shows the strongest positive correlation and the tightest clustering of data points around the regression line among all four panels. This confirms that the dominant wet snow reservoir in HTNP — residing in the broad 2000–3000 m zone — is primarily precipitation-driven, with higher snowfall inputs directly expanding the areal extent of wet snow across this vast terrain. The <1500 m band shows positive but scattered precipitation–wet snow relationships, constrained by the relatively small absolute wet snow area at these elevations. The 1500–2000 m band also shows a positive trend, though with more scatter reflecting the mixed temperature and precipitation controls at this transitional elevation. The >3000 m band shows a similarly positive trend, though wet snow occurrence above 3000 m is more limited and variable. Taken collectively, the interpretation shows that precipitation is the master variable controlling wet snow distribution in Hohe Tauren, with its influence most strongly expressed in the mid-to-high elevation bands that constitute the bulk of the park's snowpack. This has profound implications for alpine water storage: maintaining adequate winter precipitation is critical to sustaining the wet snowpack that drives spring and early summer streamflow across the park's river systems.

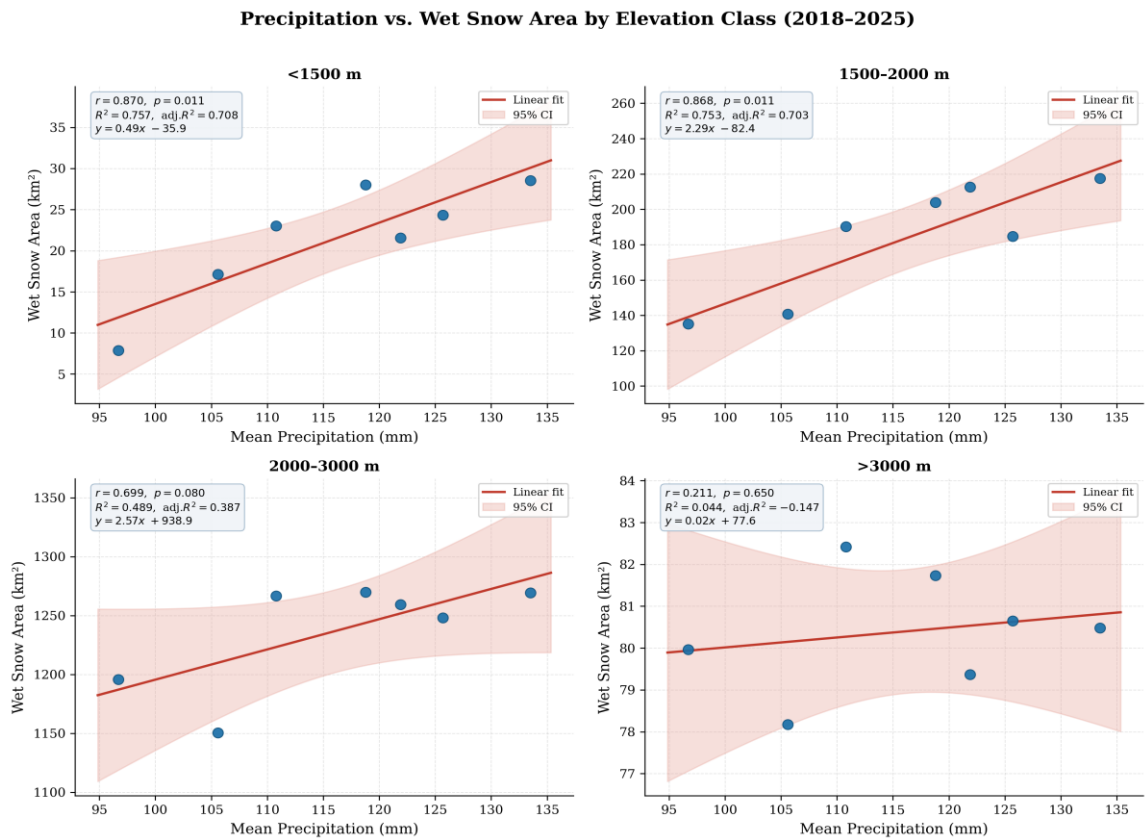


Figure 16 Four-panel scatter plots of mean accumulation-season precipitation (mm) versus wet snow area (km²) for each elevation class across the study area.

6 DISCUSSION

The results obtained in this study were discussed within the context of relevant literature, methodology, limitations, and research outlook. The discussion part is organized in the order of presented research questions and covers the performance of the chosen classifier, applicability of different sensors, identified temporal trends, climatic drivers of snow cover dynamics, and methodological aspects of the current study, including transferability.

6.1 Classification Performance and NDSI Threshold Selection

The rule-based classification algorithm performed successfully in the task of snow cover delineation. The AUC of the model using Sentinel-2 NDSI reached an excellent value of 0,98 for the period 2018-2025, which compares favorably to the results reported in the literature for optical methods in alpine terrain. For example, Poussin et al. (2023) showed that delineating Landsat-derived NDSI using a spatiotemporal threshold is highly efficient in complex topography and comparable to the current study since the AUC of their model equals 0,8321. Tong et al. (2020) trained a similar model for snow cover delineation in Austria using MODIS data and reported an AUC of 0,7028, which is once again close to the value found in this study. The performance of the model trained on Sentinel-2 data was better than the performance of the models trained on Landsat and MODIS data because of the higher spatial resolution of 10 meters, compared to 30 meters for Landsat and 500 meters for MODIS. Such a high resolution allowed the model to better account for the complexity of the terrain in the heterogeneous landscape of the study area.

A threshold value of 0.4 was used to delineate the snow cover in this study. The rationale for selecting this specific value was since the value was found to be adequate for optical delineation in multiple studies (Ma and Zhang, 2022; Tong et al, 2020). However, Ma and Zhang (2022) also report that dynamic thresholds, which are adjusted according to specific conditions in a certain area, allow for better delineation in mountainous areas, where forests are common, as they allow the model to account for the heterogeneity of partially covered slopes. Therefore, it would be advisable to try different threshold values in various areas in future study to see if the performance of the model can be improved.

6.2 Multi-Sensor Fusion and SAR-Based Snow-State Discrimination

The addition of Sentinel-1 C-band SAR backscatter as an input feature to the snow classification algorithm enabled distinguishing between the Dry Snow and Wet Snow classes, which is impossible for any optical sensor. The criteria used in this study for wet snow detection with SAR ($VV < -11$ dB for absolute backscatter and $\Delta VV < -2.0$ dB for relative backscatter) were similar to those reported in the SAR snow mapping literature. According to Nagler et al. (2016), detecting wet snow using Sentinel-1 SAR backscatter was possible at $VV < -10$ to -12 dB. Moreover, the approach suggested by Marin et al. (2020), which used relative backscatter change for determining wet snow was identical to the one used in this study. Those authors showed that using the ratio of multitemporal backscatter values was more efficient than using absolute thresholds for heterogeneous alpine landscapes, as it reduced the topography-driven bias.

As for the physics behind the detection of wet snow, the addition of liquid water into the snowpack increases the dielectric permittivity of snow, which leads to higher microwave attenuation and therefore lower backscatter values at C-band (Tsai et al., 2019; Lund et

al., 2022). This effect was confirmed by Brangers et al. (2024) who showed that temporal variations in the VV-polarized backscatter from an alpine snowpack observed from a tower C-band radar were strongly correlated with in situ liquid water content during the melting period. Finally, Carletti et al. (2025) demonstrated that temporal stacks of Sentinel-1 backscatter combined with physically based snowpack modeling could capture the detailed spatiotemporal pattern of the snowpack evolution. Such an approach, combined with radiative transfer modeling, should allow for quantitative assessment of the contribution of liquid water content to the observed SAR signal. Similarly, the fusion algorithm used in this study and based on PCA of Sentinel-2 and Sentinel-1 observations exploited the information contained in both optical and microwave observations. The high value of Pearson's correlation coefficient (R^2) between the first PCA component and NDSI indicated that SAR backscatter information had indeed added value to the snow state delineation, beyond the information provided by the optical sensor, which was expected based on the information theory considerations.

6.3 Observed Snow-Cover Trends in the Context of Alpine Climate Change

The tendency of the data on the transition from dominance of Dry Snow to Wet Snow and an increase in the non-snow area at the expense of the lower areas, as well as the lifting of the snow line, is consistent with the trends of regional warming reported in the Austrian and Alpine literature. Thus, Schöner et al. (2009) also find a reduction in snow depth at the Sonnblick Observatory (3106 m a.s.l.) during the XX century, which was mainly due to the decrease in the share of snowfall and temperature increases that accelerate snowmelt in spring. Olefs et al. (2020) confirm the recent trend of negative trends in snow depth and snow cover duration for Austria for the period 1961-2020, which were most noticeable for the lowest altitudes, and the authors' findings match the patterns found in this study. The study's results of the zone's mean LST crossing the zero level in 2025 and the non-snow area peaking at 42.49 km² in 2023 fit within the limits of rapid low-altitude snowpack decline, reported for the Austrian Alps.

The authors' projections of the resilience of high-altitude snowpack can also be supported by findings from the literature. Thus, Hanzer et al. (2018) model cryospheric changes in the Ötztal Alps under the RCP scenarios and find that while the accumulation areas at high altitudes will retain significant snowpack until mid-century, valley glaciers and low-altitude snowfields will disappear with relative rapidity under global warming. The 2-3 °C warming of the temperature at Zone IV (more than 3000 m a.s.l.) during 2018-2025 found in the current study also demonstrates that even the high-altitude snowpack is not protected from warming and will be affected after a certain period.

The snowpack recovery that has occurred in 2021, which is reflected in the higher-than-average NDSI values, the lower Non-Snow area, and an anomalous 3.47 °C temperature decrease for the study period, is due to the year-to-year weather variability affecting short-term trends. Therefore, when analyzing climate change trends, such as temperature or snow cover dynamics, it is essential to analyze multi-year trends rather than focusing on one year's results.

6.4 Climatic Drivers and Elevation Interactions

The conclusion that precipitation is the primary driver of wet snow extent in the 2000-3000 m elevation band, extracted from the analysis, appears reasonable given the presented results. The fact that the area shows the highest positive Precipitation-Wet Snow cluster and near-zero Temperature-Wet Snow cluster suggests that for the temperature range typical of this zone, temperatures are not limiting, and, thus, the

amount of precipitation contributes most to the extent of the winter wet snowpack. Such an outcome is particularly noteworthy considering the dominance of this zone in the park's annual snow area (1150–1270 km²) and its projected impact on the amount of runoff in spring and early summer. Specifically, changes in the amount of winter precipitation and its phase at this elevation may have a disproportionately large effect on the seasonal streamflow. Further increases in temperature, as projected by most climate change scenarios, will cause the temperature threshold beyond which precipitation falls as rain to shift upwards, affecting first the lowest elevation bands, converting them from a precipitation- to temperature-limited regime, and, ultimately, decreasing the total amount of runoff.

A similarly reasonable conclusion can be drawn from the results of the weak Temperature-Precipitation link ($r \sim 0.2\text{--}0.35$) presented by the SPARTACUS data. The weak link can be explained by the fact that in summer, the Eastern Alps typically exhibit a convective precipitation regime, while in autumn and the early winter, the main contribution comes from frontal systems, which have a limited temperature range. Thus, the amount of precipitation alone cannot be used to estimate the amount of snowfall on a daily basis, as demonstrated in the analysis of Olefs et al. (2020) for the Austrian Alps.

7 METHODOLOGICAL LIMITATIONS

Several methodological constraints must be considered when interpreting the results of this study. The analysis was restricted to the 2018–2025 period due to the availability of the harmonized Sentinel-2 Level-2A surface-reflectance dataset in Google Earth Engine, which achieved consistent coverage only after 2017. While climatological assessments commonly rely on observation periods of approximately 30 years, such a duration does not by itself allow anthropogenic climate forcing to be separated conclusively from natural interannual and decadal variability. Therefore, the eight-year record used in this study is suitable for examining short-term variability and recent snow-cover changes, but not for establishing definitive long-term climate trends or attributing the observed changes to anthropogenic forcing. Furthermore, the use of seasonal median compositing, while effective for mitigating cloud cover, inherently smooths out short-term variations, making it impossible to identify sub-seasonal events such as freeze-thaw cycles, rain-on-snow, or early-season snowfalls that significantly influence snowpack structure and runoff. Additionally, the classification results lack independent in-situ verification; while AUC–ROC analysis confirmed high spectral separability, the wet and dry snow classes were not tested against physical measurements from snow lysimeters or pits at the location. Finally, the current workflow requires manual intervention to determine seasonal windows and thresholds, suggesting that future operational applications within national snow monitoring networks would require the development of a fully automated, dynamic processing chain.

8 CONCLUSION

This article describes a multisensor remote sensing method for identifying, classifying, and analyzing snow cover in Hohe Tauern National Park, Austria, between 2018 and 2025. Snow was detected using Sentinel-2 optical, Sentinel-1 C-band Synthetic Aperture Radar (SAR), ALOS PALSAR Digital Terrain Model (DEM), and SPARTACUS climate reanalysis data. It was found that the use of combined optical-Radar data for snow cover detection using the Principal Component Analysis method allows obtaining better results than using individual sensors. The rule-based algorithm successfully separated Non-Snow, Dry Snow, and Wet Snow classes with an accuracy of 89%. An increase in the amount of Wet Snow was found for all altitudes, with its contribution being higher at lower altitudes, which is associated with the overall warming trend. The snowline's upward shift was also identified for all altitudinal bands. The AUC-ROC analysis showed that for Alpine snow cover detection, Sentinel-2's NDSI performs best (AUC equals 0.98) compared to Landsat 8/9 OLI (AUC – 0.83) and MODIS (AUC – 0.7). This allows using the obtained results to assess the current state of the snow cover in alpine ecosystems.

The snow cover in Hohe Tauern National Park was found to be primarily associated with elevation. High elevation areas (>3000 m) were characterized by a stable snow cover with NDSI values approaching 1. In the study period, the zones between 2000 and 3000 m tend to lose snow by 2025, while the areas below 2000 m were identified as the most affected by warming, crossing the threshold of 0°C by 2025. The amount of Dry Snow was found to depend on the slope exposure and decrease from north to south, following the difference in solar radiation receipt. At the same time, the relationship between the amount of precipitation and the share of Wet Snow was the strongest in the altitude class between 2000 and 3000 m, which is the core area of the snow cover in the park. The consistent changes in snow cover for Hohe Tauern National Park suggest a similar development of processes in other mountainous areas, affecting the water supply, ecosystems, tourism, and natural hazard regimes of the region.

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ANEX 1

S1_S2_PCA_SNOW

```
//
=====
=====
// HOHE TAUERN NATIONAL PARK — REPRODUCIBLE SNOW WORKFLOW
//
// Required:
// Import the study-area boundary and name it: table
//
// Output classes:
// 0 = Non-Snow
// 1 = Dry Snow
// 2 = Wet Snow
//
// Classification:
// Snow = NDSI > 0.4
// Non-Snow = NDSI <= 0.4
// Wet Snow = Snow AND (VV < -11 dB OR dVV <= -2 dB)
// Dry Snow = Remaining Snow pixels
//
// PCA inputs:
// B2, B3, B4, B8, B11, NDVI, VV and VH
//
// NDSI is calculated independently and is not included in PCA.
//
=====
=====

// USER SETTINGS

var aoiFc = table;
var aoi = aoiFc.geometry();

var FIRST_YEAR = 2018;
var LAST_YEAR = 2025;

// Result displayed on the map.
var DISPLAY_YEAR = 2023;
var DISPLAY_SEASON = 'accumulation';
// Use 'accumulation' or 'depletion'.

var OUTPUT_SCALE = 10;
var PCA_SCALE = 20;
var OUTPUT_CRS = 'EPSG:32633';
var MAX_PIXELS = 1e13;
var TILE_SCALE = 4;

// Thresholds.
var NDSI_THRESHOLD = 0.4;
var VV_WET_THRESHOLD_DB = -11;
var DELTA_VV_WET_THRESHOLD_DB = -2.0;

// Minimum mapping unit.
var APPLY_MMU = true;
var MIN_PATCH_PIXELS = 10;
```

```

// Export settings.
var EXPORT_FOLDER = 'Hohe-Tauern_Snow_Study';

// Keep false while testing.
// Change to true to create exports for all 16 seasonal results.
var EXPORT_ALL_SEASONS = false;

var EXPORT_NDSI = true;
var EXPORT_PCA_INDEX = true;
var EXPORT_CLASSIFICATION = true;
var EXPORT_DELTA_VV = true;
var EXPORT_CHANGE_MAPS = false;
var EXPORT_TABLES = true;

// MAP SETUP

Map.centerObject(aoiFc, 9);

Map.addLayer(
  aoiFc.style({
    color: 'red',
    fillColor: '00000000',
    width: 2
  }),
  {},
  'Study area',
  true
);

// YEARS, SEASONS AND DATE WINDOWS

var years = [];

for (var year = FIRST_YEAR; year <= LAST_YEAR; year++) {
  years.push(year);
}

var seasons = [
  'accumulation',
  'depletion'
];

// filterDate uses an exclusive ending date.
function getSeasonDates(year, season) {

  if (season === 'accumulation') {
    return {
      start: String(year - 1) + '-12-15',
      end: String(year) + '-03-01'
    };
  }

  if (season === 'depletion') {
    return {
      start: String(year) + '-06-15',
      end: String(year) + '-09-16'
    };
  }
}

```

```

    };
  }

  throw new Error('Unknown season: ' + season);
}

// SENTINEL-2 PREPROCESSING

var S2_REFLECTANCE_BANDS = [
  'B2',
  'B3',
  'B4',
  'B8',
  'B11'
];

function prepareSentinel2(image) {

  var scl = image.select('SCL');

  // SCL 3 = Cloud shadow
  // SCL 8 = Medium-probability cloud
  // SCL 9 = High-probability cloud
  // SCL 10 = Thin cirrus
  //
  // SCL 11 is Snow/Ice and is intentionally retained.

  var invalidPixels = scl.eq(3)
    .or(scl.eq(8))
    .or(scl.eq(9))
    .or(scl.eq(10));

  var reflectance = image
    .updateMask(invalidPixels.not())
    .select(S2_REFLECTANCE_BANDS)
    .resample('bilinear')
    .multiply(0.0001)
    .toFloat();

  var ndsi = reflectance
    .normalizedDifference([
      'B3',
      'B11'
    ])
    .rename('NDSI');

  var ndvi = reflectance
    .normalizedDifference([
      'B8',
      'B4'
    ])
    .rename('NDVI');

  return reflectance
    .addBands([
      ndsi,
      ndvi
    ])

```

```

        .copyProperties(
            image,
            [
                'system:time_start',
                'system:index'
            ]
        );
    }
}

```

```

function getSentinel2Collection(startDate, endDate) {

    return ee.ImageCollection(
        'COPERNICUS/S2_SR_HARMONIZED'
    )
    .filterBounds(aoi)
    .filterDate(startDate, endDate)
    .filter(
        ee.Filter.lt(
            'CLOUDY_PIXEL_PERCENTAGE',
            80
        )
    )
    .map(prepareSentinel2);
}

```

```

function getSentinel2Composite(startDate, endDate) {

    var collection = getSentinel2Collection(
        startDate,
        endDate
    );

    return {
        collection: collection,

        count: collection.size(),

        image: collection
            .median()
            .clip(aoi)
            .toFloat()
    };
}

```

```

// SENTINEL-1 PREPROCESSING

```

```

// COPERNICUS/S1_GRD is already provided in decibels.
// Do not perform another 10 × log10 conversion.

```

```

function getSentinel1Collection(startDate, endDate) {

    return ee.ImageCollection(
        'COPERNICUS/S1_GRD'
    )
    .filterBounds(aoi)
    .filterDate(startDate, endDate)
}

```



```

.filter(
  ee.Filter.eq(
    'instrumentMode',
    'IW'
  )
)
.filter(
  ee.Filter.eq(
    'orbitProperties_pass',
    'ASCENDING'
  )
)
.filter(
  ee.Filter.listContains(
    'transmitterReceiverPolarisation',
    'VV'
  )
)
.filter(
  ee.Filter.listContains(
    'transmitterReceiverPolarisation',
    'VH'
  )
)
.select([
  'VV',
  'VH'
]);
}

```

```

function getSentinel1Composite(startDate, endDate) {

  var collection = getSentinel1Collection(
    startDate,
    endDate
  );

  return {
    collection: collection,

    count: collection.size(),

    image: collection
      .median()
      .clip(aoi)
      .toFloat()
  };
}

```

```
// MULTI-YEAR SENTINEL-1 REFERENCE
```

```

var baselineCollection = ee.ImageCollection([]);

years.forEach(function(yearValue) {

  var dates = getSeasonDates(
    yearValue,

```

```

    'accumulation'
  );

  baselineCollection = baselineCollection.merge(
    getSentinel1Collection(
      dates.start,
      dates.end
    )
  );
});

var vvReference = baselineCollection
  .select('VV')
  .median()
  .rename('VV_reference_dB')
  .clip(aoi)
  .toFloat();

print(
  'Number of Sentinel-1 images in VV reference:',
  baselineCollection.size()
);

// PCA SETTINGS

var PCA_BANDS = [
  'B2',
  'B3',
  'B4',
  'B8',
  'B11',
  'NDVI',
  'VV',
  'VH'
];

var NUMBER_OF_PCA_BANDS = PCA_BANDS.length;

var PC_NAMES = [];
var SD_NAMES = [];

for (
  var component = 1;
  component <= NUMBER_OF_PCA_BANDS;
  component++
) {

  PC_NAMES.push(
    'PC' + component
  );

  SD_NAMES.push(
    'SD' + component
  );
}

```

```
// Z-SCORE STANDARDIZATION

function standardizeImage(
  image,
  region,
  analysisScale
) {

  var bandNames = image.bandNames();

  var statistics = image.reduceRegion({

    reducer: ee.Reducer.mean().combine({
      reducer2: ee.Reducer.stdDev(),
      sharedInputs: true
    }),

    geometry: region,

    scale: analysisScale,

    maxPixels: MAX_PIXELS,

    tileScale: TILE_SCALE
  });

  var means = ee.Image.constant(

    bandNames.map(function(bandName) {

      bandName = ee.String(bandName);

      return ee.Number(
        statistics.get(
          bandName.cat('_mean')
        )
      );
    })

  ).rename(bandNames);

  var standardDeviations = ee.Image.constant(

    bandNames.map(function(bandName) {

      bandName = ee.String(bandName);

      return ee.Number(
        statistics.get(
          bandName.cat('_stdDev')
        )
      ).max(1e-6);
    })

  ).rename(bandNames);
}
```

```

return image
  .subtract(means)
  .divide(standardDeviations)
  .rename(bandNames)
  .toFloat();
}

// PCA FUNCTION

function runPCA(
  standardizedImage,
  region,
  analysisScale
) {

  var arrayImage = standardizedImage.toArray();

  var covarianceDictionary = arrayImage.reduceRegion({

    reducer: ee.Reducer.centeredCovariance(),

    geometry: region,

    scale: analysisScale,

    maxPixels: MAX_PIXELS,

    tileScale: TILE_SCALE
  });

  var covarianceArray = ee.Array(
    covarianceDictionary.get('array')
  );

  var eigenPairs = covarianceArray.eigen();

  // First column contains eigenvalues.
  var eigenValues = eigenPairs.slice(
    1,
    0,
    1
  );

  // Remaining columns contain eigenvectors.
  var eigenVectors = eigenPairs.slice(
    1,
    1
  );

  var arrayImage2D = arrayImage.toArray(1);

```

```

var principalComponents = ee.Image(
  eigenVectors
)
  .matrixMultiply(arrayImage2D)
  .arrayProject([0])
  .arrayFlatten([PC_NAMES]);

var standardDeviationImage = ee.Image(
  eigenValues.sqrt()
)
  .arrayProject([0])
  .arrayFlatten([SD_NAMES]);

var standardizedPCs = principalComponents
  .divide(standardDeviationImage)
  .rename(PC_NAMES)
  .clip(region)
  .toFloat();

return {
  pcImage: standardizedPCs,
  eigenValues: eigenValues,
  eigenVectors: eigenVectors
};
}

// CORRELATE PRINCIPAL COMPONENTS WITH NDSI

function getPCCorrelationTable(
  pcImage,
  ndsiImage,
  yearValue,
  seasonName
) {

  var features = PC_NAMES.map(
    function(pcName) {

      var correlationDictionary = pcImage
        .select(pcName)
        .addBands(
          ndsiImage.rename('NDSI')
        )
        .reduceRegion({

          reducer:
            ee.Reducer.pearsonsCorrelation(),

          geometry: aoi,

          scale: PCA_SCALE,

          maxPixels: MAX_PIXELS,

```

```

        tileScale: TILE_SCALE
    });

    var correlation = ee.Number(

        ee.Algorithms.If(

            ee.Dictionary(
                correlationDictionary
            ).contains('correlation'),

            correlationDictionary.get(
                'correlation'
            ),

            0
        )
    );

    return ee.Feature(null, {

        year: yearValue,

        season: seasonName,

        principal_component: pcName,

        correlation_r: correlation,

        absolute_correlation:
            correlation.abs()
    });
}
);

return ee.FeatureCollection(features);
}

// MINIMUM MAPPING UNIT

function applyMinimumMappingUnit(classImage) {

    if (!APPLY_MMU) {
        return classImage;
    }

    var localMajority = classImage.focalMode(
        1,
        'square',
        'pixels',
        1
    );
};

```

```

var cleaned = classImage;

[1, 2].forEach(function(classValue) {

    var classMask = classImage.eq(
        classValue
    );

    var connectedCount = classMask
        .selfMask()
        .connectedPixelCount(
            256,
            true
        );

    var smallPatch = classMask.and(
        connectedCount.lt(
            MIN_PATCH_PIXELS
        )
    );

    cleaned = cleaned.where(
        smallPatch,
        localMajority
    );
});

return cleaned
    .rename('Snow_Class')
    .updateMask(classImage.mask())
    .toByte();
}

// CLASS AREA IN SQUARE KILOMETRES

function getClassAreaKm2(
    classImage,
    classValue
) {

    var areaDictionary = ee.Image
        .pixelArea()
        .divide(1e6)
        .rename('area_km2')
        .updateMask(
            classImage.eq(classValue)
        )
        .reduceRegion({

            reducer: ee.Reducer.sum(),

            geometry: aoi,

```

```

        scale: OUTPUT_SCALE,

        crs: OUTPUT_CRS,

        maxPixels: MAX_PIXELS,

        tileScale: TILE_SCALE
    });

    return ee.Number(

        ee.Algorithms.If(

            areaDictionary.contains(
                'area_km2'
            ),

            areaDictionary.get(
                'area_km2'
            ),

            0
        )
    );
}

// BUILD ONE SEASONAL RESULT

function buildSeasonResult(
    yearValue,
    seasonName
) {

    var dates = getSeasonDates(
        yearValue,
        seasonName
    );

    var s2Result = getSentinel2Composite(
        dates.start,
        dates.end
    );

    var s1Result = getSentinel1Composite(
        dates.start,
        dates.end
    );

    var s2Composite = s2Result.image;
    var s1Composite = s1Result.image;

    var ndsi = s2Composite.select(
        'NDSI'
    );

```



```

);

var deltaVV = s1Composite
  .select('VV')
  .subtract(vvReference)
  .rename('dVV')
  .toFloat();

// Valid optical and radar observations.
var validDataMask = ndsi.mask()
  .and(
    s1Composite
      .select('VV')
      .mask()
  )
  .and(
    s1Composite
      .select('VH')
      .mask()
  )
  .and(
    deltaVV.mask()
  );

// Eight-variable PCA input stack.
var pcaStack = s2Composite
  .select([
    'B2',
    'B3',
    'B4',
    'B8',
    'B11',
    'NDVI'
  ])
  .addBands(
    s1Composite.select([
      'VV',
      'VH'
    ])
  )
  .select(PCA_BANDS)
  .updateMask(validDataMask)
  .clip(aoi)
  .toFloat();

var standardizedStack = standardizeImage(
  pcaStack,
  aoi,
  PCA_SCALE
);

var pcaResult = runPCA(
  standardizedStack,
  aoi,

```

```

PCA_SCALE
);

var pcImage = pcaResult.pcImage;

// NDSI is used independently to identify
// the strongest snow-correlated PC.
var correlationTable =
  getPCCorrelationTable(
    pcImage,
    ndsi.updateMask(
      validDataMask
    ),
    yearValue,
    seasonName
  );

var bestPCFeature = ee.Feature(

  correlationTable
    .sort(
      'absolute_correlation',
      false
    )
    .first()
);

var bestPCName = ee.String(

  bestPCFeature.get(
    'principal_component'
  )
);

var bestCorrelation = ee.Number(

  bestPCFeature.get(
    'correlation_r'
  )
);

// Reverse the PC sign when required,
// so higher PC values correspond to higher NDSI.
var orientation = ee.Number(

  ee.Algorithms.If(
    bestCorrelation.lt(0),
    -1,
    1
  )
);

```

```

var pcaSnowIndex = pcImage
    .select(bestPCName)
    .multiply(orientation)
    .rename('PCA_Snow_Index')
    .updateMask(validDataMask)
    .toFloat();

// RULE-BASED CLASSIFICATION

var isSnow = ndsi.gt(
    NDSI_THRESHOLD
);

var wetRuleAbsolute = s1Composite
    .select('VV')
    .lt(
        VV_WET_THRESHOLD_DB
    );

var wetRuleRelative = deltaVV
    .lte(
        DELTA_VV_WET_THRESHOLD_DB
    );

var wetRule = wetRuleAbsolute.or(
    wetRuleRelative
);

var wetSnow = isSnow
    .and(wetRule)
    .and(validDataMask);

var drySnow = isSnow
    .and(wetRule.not())
    .and(validDataMask);

// 0 = Non-Snow
// 1 = Dry Snow
// 2 = Wet Snow
var classImage = ndsi
    .multiply(0)
    .toByte()
    .rename('Snow_Class')
    .where(
        drySnow,
        1
    )
    .where(
        wetSnow,
        2
    )
    .updateMask(validDataMask)

```

```

.clip(aoi);

var cleanedClassImage =
  applyMinimumMappingUnit(
    classImage
  );

var summaryFeature = ee.Feature(
  null,
  {

    year: yearValue,

    season: seasonName,

    start_date: dates.start,

    end_date_exclusive:
      dates.end,

    sentinel2_image_count:
      s2Result.count,

    sentinel1_image_count:
      s1Result.count,

    selected_principal_component:
      bestPCName,

    selected_pc_ndsi_correlation:
      bestCorrelation,

    non_snow_area_km2:
      getClassAreaKm2(
        cleanedClassImage,
        0
      ),

    dry_snow_area_km2:
      getClassAreaKm2(
        cleanedClassImage,
        1
      ),

    wet_snow_area_km2:
      getClassAreaKm2(
        cleanedClassImage,
        2
      )
  }
);

return {

  year: yearValue,

```

```

    season: seasonName,

    dates: dates,

    s2Composite: s2Composite,

    s1Composite: s1Composite,

    ndsi: ndsi.updateMask(
        validDataMask
    ),

    deltaVV: deltaVV.updateMask(
        validDataMask
    ),

    standardizedStack:
        standardizedStack,

    pcImage: pcImage,

    pcaSnowIndex:
        pcaSnowIndex,

    bestPCName:
        bestPCName,

    bestCorrelation:
        bestCorrelation,

    correlationTable:
        correlationTable,

    classImage:
        cleanedClassImage,

    summaryFeature:
        summaryFeature
    };
}

// PROCESS ALL YEARS AND SEASONS

var results = [];

years.forEach(function(yearValue) {

    seasons.forEach(function(seasonName) {

        results.push(

            buildSeasonResult(
                yearValue,
                seasonName
            )
        );
    });
});

```

```

function findResult(
  yearValue,
  seasonName
) {

  for (
    var index = 0;
    index < results.length;
    index++
  ) {

    if (
      results[index].year === yearValue &&
      results[index].season === seasonName
    ) {

      return results[index];
    }
  }

  throw new Error(
    'No result found for ' +
    yearValue +
    ' ' +
    seasonName
  );
}

var displayedResult = findResult(
  DISPLAY_YEAR,
  DISPLAY_SEASON
);

// SUMMARY TABLES

var summaryFeatures = results.map(
  function(result) {

    return result.summaryFeature;
  }
);

var summaryTable = ee.FeatureCollection(
  summaryFeatures
);

var allCorrelationTables =
  ee.FeatureCollection([]);

results.forEach(function(result) {

```

```

allCorrelationTables =
  allCorrelationTables.merge(
    result.correlationTable
  );
});

print(
  'Seasonal summary table:',
  summaryTable
);

print(
  'PCA-NDSI correlations for displayed result:',

  displayedResult
    .correlationTable
    .sort(
      'absolute_correlation',
      false
    )
);

print(
  'Selected PC:',
  displayedResult.bestPCName
);

print(
  'Selected PC correlation with NDSI:',
  displayedResult.bestCorrelation
);

// INTERANNUAL SNOW-COVER CHANGE

// 0 = Persistent Non-Snow
// 1 = Snow loss
// 2 = Snow gain
// 3 = Persistent Snow

function buildChangeResult(
  previousResult,
  currentResult
) {

  var previousSnow =
    previousResult.classImage.gt(0);

  var currentSnow =
    currentResult.classImage.gt(0);

  var commonMask =
    previousResult.classImage.mask()

```

```

        .and(
            currentResult.classImage.mask()
        );

var changeImage = previousSnow
    .multiply(0)
    .toByte()
    .rename('Snow_Change')
    .where(
        previousSnow.and(
            currentSnow.not()
        ),
        1
    )
    .where(
        previousSnow.not().and(
            currentSnow
        ),
        2
    )
    .where(
        previousSnow.and(
            currentSnow
        ),
        3
    )
    .updateMask(commonMask)
    .clip(aoi);

return {

    previousYear:
        previousResult.year,

    currentYear:
        currentResult.year,

    season:
        currentResult.season,

    image:
        changeImage
};
}

var changeResults = [];

seasons.forEach(function(seasonName) {

    for (
        var changeYear = FIRST_YEAR + 1;
        changeYear <= LAST_YEAR;
        changeYear++
    ) {

```



```

changeResults.push(
    buildChangeResult(
        findResult(
            changeYear - 1,
            seasonName
        ),
        findResult(
            changeYear,
            seasonName
        )
    )
);
}
});

```

// VISUALIZATION

```

var rgbVis = {
    bands: [
        'B4',
        'B3',
        'B2'
    ],
    min: 0.0,
    max: 0.35
};

```

```

var ndsiVis = {
    min: -0.5,
    max: 1.0,
    palette: [
        '8c510a',
        'f6e8c3',
        'ffffff',
        '67a9cf',
        '2166ac'
    ]
};

```

```

var pcaVis = {
    min: -3,
    max: 3,
    palette: [
        '8c510a',
        'f6e8c3',
        'ffffff',
        '67a9cf',
        '2166ac'
    ]
};

```

```

var deltaVVVis = {

```

```

min: -6,
max: 6,
palette: [
  'b2182b',
  'f7f7f7',
  '2166ac'
]
};

```

```

var snowClassVis = {
  min: 0,
  max: 2,
  palette: [
    '8c6d31',
    'ffffff',
    '00bcd4'
  ]
};

```

```

Map.addLayer(
  displayedResult.s2Composite,
  rgbVis,
  DISPLAY_YEAR +
  ' ' +
  DISPLAY_SEASON +
  ' Sentinel-2 RGB',
  false
);

```

```

Map.addLayer(
  displayedResult.ndsi,
  ndsiVis,
  DISPLAY_YEAR +
  ' ' +
  DISPLAY_SEASON +
  ' NDSI',
  false
);

```

```

Map.addLayer(
  displayedResult.pcaSnowIndex,
  pcaVis,
  DISPLAY_YEAR +
  ' ' +
  DISPLAY_SEASON +
  ' snow-correlated PC',
  false
);

```

```

Map.addLayer(
  displayedResult
    .s1Composite
    .select('VV'),

```

```

    {
      min: -25,
      max: 0
    },

    DISPLAY_YEAR +
    ' ' +
    DISPLAY_SEASON +
    ' Sentinel-1 VV (dB)',

    false
  );

  Map.addLayer(
    displayedResult.deltaVV,
    deltaVVVis,
    DISPLAY_YEAR +
    ' ' +
    DISPLAY_SEASON +
    ' dVV (dB)',
    false
  );

  Map.addLayer(
    displayedResult.classImage,
    snowClassVis,
    DISPLAY_YEAR +
    ' ' +
    DISPLAY_SEASON +
    ' Snow classification',
    true
  );

  // LEGEND

  function addLegendRow(
    panel,
    colour,
    label
  ) {

    var colourBox = ui.Label({

      style: {

        backgroundColor:
          '#' + colour,

        padding:
          '8px',

        margin:
          '0 8px 4px 0'
      }
    });
  }

```

```

var textLabel = ui.Label({

  value:
    label,

  style: {
    margin:
      '0 0 4px 0'
  }
});

panel.add(

  ui.Panel({

    widgets: [
      colourBox,
      textLabel
    ],

    layout:
      ui.Panel.Layout.Flow(
        'horizontal'
      )
  })
);
}

```

```

var legend = ui.Panel({

  style: {

    position:
      'bottom-left',

    padding:
      '8px 12px'
  }
});

```

```

legend.add(

  ui.Label({

    value:
      'Snow classification',

    style: {

      fontWeight:
        'bold',

      margin:
        '0 0 6px 0'
    }
  })
);

```

```
    })  
  );
```

```
addLegendRow(  
  legend,  
  '8c6d31',  
  'Non-Snow'  
);
```

```
addLegendRow(  
  legend,  
  'ffffff',  
  'Dry Snow'  
);
```

```
addLegendRow(  
  legend,  
  '00bcd4',  
  'Wet Snow'  
);
```

```
Map.add(legend);
```

```
// EXPORT FUNCTION
```

```
function exportSeasonResult(result) {
```

```
  var prefix =  
    String(result.year) +  
    '_' +  
    result.season;
```

```
  if (EXPORT_NDSI) {
```

```
    Export.image.toDrive({
```

```
      image:  
        result.ndsi  
          .unmask(-9999)  
          .toFloat(),
```

```
      description:  
        'NDSI_' + prefix,
```

```
      folder:  
        EXPORT_FOLDER,
```

```
      fileNamePrefix:  
        'ndsi_' + prefix,
```

```
      region:  
        aoi,
```

```

scale:
  OUTPUT_SCALE,

crs:
  OUTPUT_CRS,

maxPixels:
  MAX_PIXELS,

fileFormat:
  'GeoTIFF',

formatOptions: {
  noData: -9999
}
});
}

if (EXPORT_PCA_INDEX) {

  Export.image.toDrive({

    image:
      result.pcaSnowIndex
        .unmask(-9999)
        .toFloat(),

    description:
      'PCA_Snow_Index_' +
      prefix,

    folder:
      EXPORT_FOLDER,

    fileNamePrefix:
      'pca_snow_index_' +
      prefix,

    region:
      aoi,

    scale:
      OUTPUT_SCALE,

    crs:
      OUTPUT_CRS,

    maxPixels:
      MAX_PIXELS,

    fileFormat:
      'GeoTIFF',

    formatOptions: {
      noData: -9999
    }
  });
}

```

```
if (EXPORT_CLASSIFICATION) {
```

```
    Export.image.toDrive({
```

```
        image:
            result.classImage
                .unmask(255)
                .toUint8(),
```

```
        description:
            'Snow_Class_' +
            prefix,
```

```
        folder:
            EXPORT_FOLDER,
```

```
        fileNamePrefix:
            'snow_class_' +
            prefix,
```

```
        region:
            aoi,
```

```
        scale:
            OUTPUT_SCALE,
```

```
        crs:
            OUTPUT_CRS,
```

```
        maxPixels:
            MAX_PIXELS,
```

```
        fileFormat:
            'GeoTIFF',
```

```
        formatOptions: {
            noData: 255
        }
```

```
    });
```

```
}
```

```
if (EXPORT_DELTA_VV) {
```

```
    Export.image.toDrive({
```

```
        image:
            result.deltaVV
                .unmask(-9999)
                .toFloat(),
```

```
        description:
            'Delta_VV_' +
            prefix,
```

```
        folder:
            EXPORT_FOLDER,
```

```

    fileNamePrefix:
      'delta_vv_' +
      prefix,

    region:
      aoi,

    scale:
      OUTPUT_SCALE,

    crs:
      OUTPUT_CRS,

    maxPixels:
      MAX_PIXELS,

    fileFormat:
      'GeoTIFF',

    formatOptions: {
      noData: -9999
    }
  });
}
}

```

```
// EXPORT SELECTED OR ALL SEASONS
```

```

if (EXPORT_ALL_SEASONS) {

  results.forEach(function(result) {

    exportSeasonResult(result);
  });

} else {

  exportSeasonResult(
    displayedResult
  );
}

```

```
// OPTIONAL CHANGE-MAP EXPORTS
```

```

if (EXPORT_CHANGE_MAPS) {

  changeResults.forEach(
    function(changeResult) {

      var prefix =
        String(
          changeResult.previousYear
        ) +
        '_' +
        String(
          changeResult.currentYear

```



```

    ) +
    '_' +
    changeResult.season;

Export.image.toDrive({

    image:
        changeResult.image
        .unmask(255)
        .toUint8(),

    description:
        'Snow_Change_' +
        prefix,

    folder:
        EXPORT_FOLDER,

    fileNamePrefix:
        'snow_change_' +
        prefix,

    region:
        aoi,

    scale:
        OUTPUT_SCALE,

    crs:
        OUTPUT_CRS,

    maxPixels:
        MAX_PIXELS,

    fileFormat:
        'GeoTIFF',

    formatOptions: {
        noData: 255
    }
});
}
);
}

```

```

// TABLE EXPORTS

```

```

if (EXPORT_TABLES) {

    Export.table.toDrive({

        collection:
            summaryTable,

        description:
            'Seasonal_Snow_Area_Summary_2018_2025',
    }
}

```

```

folder:
  EXPORT_FOLDER,

fileNamePrefix:
  'seasonal_snow_area_summary_2018_2025',

fileFormat:
  'CSV'
});

```

```

Export.table.toDrive({

  collection:
    allCorrelationTables,

  description:
    'PCA_NDSI_Correlations_2018_2025',

  folder:
    EXPORT_FOLDER,

  fileNamePrefix:
    'pca_ndsi_correlations_2018_2025',

  fileFormat:
    'CSV'
});
}

```

// FINAL INFORMATION

```

print(
  'Classification rules:',

  'Non-Snow: NDSI <= ' +
    NDSI_THRESHOLD +

  '; Dry Snow: NDSI > ' +
    NDSI_THRESHOLD +
    ' and no wet-snow rule is satisfied' +

  '; Wet Snow: NDSI > ' +
    NDSI_THRESHOLD +

  ' and (VV < ' +
    VV_WET_THRESHOLD_DB +

  ' dB OR dVV <= ' +
    DELTA_VV_WET_THRESHOLD_DB +
    ' dB).'
);

print(
  'Export instruction:',
  'Open the Tasks tab and click Run for each generated export task.'
);

```

S1_S2_DRY_WET_SNOW_CLASSIFICATION

/*****

HOHE TAUERN NATIONAL PARK

RULE-BASED NON-SNOW / DRY SNOW / WET SNOW CLASSIFICATION

Thesis method:

0 = Non-Snow

1 = Dry Snow

2 = Wet Snow

Snow extent:

Snow = NDSI > 0.4

Non-Snow = NDSI <= 0.4

Wet-snow classification:

Wet Snow = Snow AND (VV < -11 dB OR deltaVV <= -2.0 dB)

Dry-snow classification:

Dry Snow = Remaining NDSI-classified snow pixels

Processing:

- Sentinel-2 Level-2A seasonal median composite
- SCL classes 3, 8, 9 and 10 masked
- SCL class 11 retained because it represents Snow/Ice
- Sentinel-1 IW ascending VV and VH observations
- Sentinel-1 GRD used directly in decibels
- Multi-year accumulation-season VV reference for 2018–2025
- Ten-pixel minimum mapping unit
- WGS 84 / UTM Zone 33N export: EPSG:32633

REQUIRED:

Import the Hohe Tauern study-area boundary and name it: table

*****/

//

=====

=====

// 0. USER SETTINGS

//

=====

=====

var aoiFc = table;

var aoi = aoiFc.geometry();

// Select the seasonal result to classify.

var analysisYear = 2023;

var analysisSeason = 'depletion';

```

// Allowed values:
// 'accumulation'
// 'depletion'

// Study period used to create the multi-year Sentinel-1 reference.

var firstReferenceYear = 2018;
var lastReferenceYear = 2025;

// Thresholds used in the thesis.

var ndsiSnowThreshold = 0.4;

var vvAbsoluteThreshold = -11;

var deltaVVThreshold = -2.0;

// Processing and export settings.

var processingScale = 10;

var exportCrs = 'EPSG:32633';

var minSnowAreaPixels = 10;

var maxPixels = 1e13;

var tileScale = 4;

// Google Drive export folder.

var exportFolder = 'Hohe-Tauren Snow Study';

//
=====
// 1. MAP SETUP
//
=====

Map.centerObject(aoiFc, 9);

Map.addLayer(
  aoiFc.style({
    color: 'red',
    fillColor: '00000000',
    width: 2
  }),
  {},
  'Study area',
  true
);

```

```

//
=====
====
// 2. SEASONAL DATE WINDOWS
//
// Earth Engine filterDate() uses:
// start date = inclusive
// end date  = exclusive
//
// Accumulation year Y:
// 15 December Y-1 to the end of February Y
//
// Depletion year Y:
// 15 June to 15 September Y
//
=====
====

function getSeasonDates(year, season) {

  if (season === 'accumulation') {

    return {
      start: String(year - 1) + '-12-15',
      end: String(year) + '-03-01'
    };
  }

  if (season === 'depletion') {

    return {
      start: String(year) + '-06-15',
      end: String(year) + '-09-16'
    };
  }

  throw new Error(
    "analysisSeason must be either 'accumulation' or 'depletion'."
  );
}

var targetDates = getSeasonDates(
  analysisYear,
  analysisSeason
);

print('Analysis year:', analysisYear);

print('Analysis season:', analysisSeason);

print('Target start date:', targetDates.start);

print(

```

```

    'Target end date, exclusive:',
    targetDates.end
);

//
=====
====
// 3. SENTINEL-2 LEVEL-2A PREPROCESSING
//
// SCL classes removed:
//
// 3 = Cloud Shadows
// 8 = Clouds Medium Probability
// 9 = Clouds High Probability
// 10 = Cirrus
//
// SCL class 11 = Snow/Ice and is deliberately retained.
//
=====
====

function prepareSentinel2(image) {

    var scl = image.select('SCL');

    var invalidPixels = scl.eq(3)
        .or(scl.eq(8))
        .or(scl.eq(9))
        .or(scl.eq(10));

    var reflectance = image
        .updateMask(invalidPixels.not())
        .select([
            'B2',
            'B3',
            'B4',
            'B8',
            'B11'
        ])
        .resample('bilinear')
        .multiply(0.0001)
        .toFloat();

    var ndsi = reflectance
        .normalizedDifference([
            'B3',
            'B11'
        ])
        .rename('NDSI');

    var ndvi = reflectance
        .normalizedDifference([
            'B8',
            'B4'
        ])

```

```

    ])
    .rename('NDVI');

return reflectance
    .addBands([
        ndsi,
        ndvi
    ])
    .copyProperties(
        image,
        [
            'system:time_start',
            'system:index'
        ]
    );
}

var sentinel2Collection = ee.ImageCollection(
    'COPERNICUS/S2_SR_HARMONIZED'
)
    .filterBounds(aoi)
    .filterDate(
        targetDates.start,
        targetDates.end
    )
    .filter(
        ee.Filter.lt(
            'CLOUDY_PIXEL_PERCENTAGE',
            80
        )
    )
    .map(prepareSentinel2);

print(
    'Sentinel-2 image count:',
    sentinel2Collection.size()
);

var sentinel2Composite = sentinel2Collection
    .median()
    .clip(aoi)
    .toFloat();

var ndsiImage = sentinel2Composite
    .select('NDSI');

Map.addLayer(
    sentinel2Composite.select([
        'B4',
        'B3',
        'B2'
    ]),
    {

```

```

        min: 0,
        max: 0.35
    },
    'Sentinel-2 RGB seasonal median',
    false
);

```

```

Map.addLayer(
    ndsiImage,
    {
        min: -0.5,
        max: 1.0,
        palette: [
            '8c510a',
            'f6e8c3',
            'ffffff',
            '67a9cf',
            '2166ac'
        ]
    },
    'NDSI seasonal median',
    false
);

```

```

//
=====
====
// 4. SENTINEL-1 GRD PREPROCESSING
//
// COPENNICUS/S1_GRD values are already provided in decibels.
// Do not apply another 10 × log10 conversion.
//
=====
=====

```

```

function getSentinel1Collection(
    startDate,
    endDate
) {

    return ee.ImageCollection(
        'COPENNICUS/S1_GRD'
    )
        .filterBounds(aoi)
        .filterDate(
            startDate,
            endDate
        )
        .filter(
            ee.Filter.eq(
                'instrumentMode',
                'IW'
            )
        )
        .filter(
            ee.Filter.eq(
                'orbitProperties_pass',

```



```

        'ASCENDING'
    )
)
.filter(
    ee.Filter.eq(
        'resolution_meters',
        10
    )
)
.filter(
    ee.Filter.listContains(
        'transmitterReceiverPolarisation',
        'VV'
    )
)
.filter(
    ee.Filter.listContains(
        'transmitterReceiverPolarisation',
        'VH'
    )
)
.select([
    'VV',
    'VH'
]);
}

```

```

// Target-season Sentinel-1 observations.

```

```

var sentinel1TargetCollection = getSentinel1Collection(
    targetDates.start,
    targetDates.end
);

```

```

print(
    'Sentinel-1 target image count:',
    sentinel1TargetCollection.size()
);

```

```

var sentinel1Target = sentinel1TargetCollection
    .median()
    .clip(aoi)
    .toFloat();

```

```

Map.addLayer(
    sentinel1Target.select('VV'),
    {
        min: -25,
        max: 0
    },
    'Sentinel-1 target VV, dB',
    false
);

```

```

Map.addLayer(
  sentinel1Target.select('VH'),
  {
    min: -32,
    max: -5
  },
  'Sentinel-1 target VH, dB',
  false
);

//
=====
====
// 5. MULTI-YEAR VV REFERENCE BASELINE
//
// The reference is the median of all ascending Sentinel-1 observations
// acquired during the accumulation windows from 2018 to 2025.
//
=====
====

var sentinel1ReferenceCollection = ee.ImageCollection([]);

for (
  var referenceYear = firstReferenceYear;
  referenceYear <= lastReferenceYear;
  referenceYear++
) {

  var referenceDates = getSeasonDates(
    referenceYear,
    'accumulation'
  );

  sentinel1ReferenceCollection =
    sentinel1ReferenceCollection.merge(

      getSentinel1Collection(
        referenceDates.start,
        referenceDates.end
      )
    );
}

print(
  'Sentinel-1 images in the multi-year reference:',
  sentinel1ReferenceCollection.size()
);

var sentinel1Reference = sentinel1ReferenceCollection
  .median()
  .clip(aoi)
  .toFloat();

```

```

var vvReference = sentinel1Reference
  .select('VV')
  .rename('VV_reference');

Map.addLayer(
  vvReference,
  {
    min: -25,
    max: 0
  },
  'Multi-year accumulation VV reference, dB',
  false
);

//
=====
// 6. RELATIVE VV CHANGE
//
// deltaVV = target VV - reference VV
//
// Negative values indicate reduced backscatter compared with the reference.
//
=====

var deltaVV = sentinel1Target
  .select('VV')
  .subtract(vvReference)
  .rename('dVV')
  .toFloat();

Map.addLayer(
  deltaVV,
  {
    min: -6,
    max: 6,
    palette: [
      'b2182b',
      'f7f7f7',
      '2166ac'
    ]
  },
  'Delta VV: target minus reference, dB',
  false
);

//
=====
// 7. RULE-BASED SNOW CLASSIFICATION
//
// NDSI > 0.4          = Snow
// NDSI <= 0.4         = Non-Snow

```

```

//
// Snow AND VV < -11 dB      = Wet Snow
// Snow AND deltaVV <= -2 dB = Wet Snow
//
// Remaining Snow           = Dry Snow
//
=====
====

// Use only pixels with valid Sentinel-2,
// target Sentinel-1 and reference Sentinel-1 observations.

var validDataMask = ndsiImage.mask()
  .and(
    sentinel1Target
      .select('VV')
      .mask()
  )
  .and(
    sentinel1Target
      .select('VH')
      .mask()
  )
  .and(
    vvReference.mask()
  )
  .and(
    deltaVV.mask()
  );

// Snow extent from Sentinel-2 NDSI.

var isSnow = ndsiImage.gt(
  ndsiSnowThreshold
);

// Absolute Sentinel-1 wet-snow rule.

var wetRuleAbsolute = sentinel1Target
  .select('VV')
  .lt(
    vvAbsoluteThreshold
  );

// Relative Sentinel-1 wet-snow rule.

var wetRuleRelative = deltaVV
  .lte(
    deltaVVThreshold
  );

// A snow pixel is Wet Snow when either
// the absolute rule or relative rule is satisfied.

```

```

var wetRule = wetRuleAbsolute.or(
  wetRuleRelative
);

var wetSnow = isSnow
  .and(wetRule)
  .and(validDataMask);

var drySnow = isSnow
  .and(wetRule.not())
  .and(validDataMask);

// Use NDSI as the base so the output inherits
// the Sentinel-2 spatial projection.

var classImage = ndsiImage
  .multiply(0)
  .rename('Snow_Class')
  .toByte()
  .where(
    drySnow,
    1
  )
  .where(
    wetSnow,
    2
  )
  .updateMask(validDataMask)
  .clip(aoi);

// Diagnostic layers.

Map.addLayer(
  isSnow
    .updateMask(isSnow)
    .updateMask(validDataMask),
  {
    palette: ['00bcd4']
  },
  'NDSI snow mask',
  false
);

Map.addLayer(
  wetRuleAbsolute
    .updateMask(wetRuleAbsolute)
    .updateMask(validDataMask),
  {
    palette: ['ff00ff']
  },
  'Wet rule: VV < -11 dB',
  false
);

```

```

Map.addLayer(
  wetRuleRelative
    .updateMask(wetRuleRelative)
    .updateMask(validDataMask),
  {
    palette: ['ff9800']
  },
  'Wet rule: deltaVV <= -2 dB',
  false
);

```

```

//
=====
====
// 8. TEN-PIXEL MINIMUM MAPPING UNIT
//
// Small connected Dry Snow and Wet Snow patches are reassigned to
// the dominant class in the surrounding 3 × 3 neighbourhood.
//
=====
====

```

```

function applyMinimumMappingUnit(
  inputClassImage
) {

  var localMajority = inputClassImage
    .focalMode(
      1,
      'square',
      'pixels',
      1
    );

  var cleaned = inputClassImage;

  [
    1,
    2
  ].forEach(function(classValue) {

    var classMask = inputClassImage.eq(
      classValue
    );

    var connectedPixels = classMask
      .selfMask()
      .connectedPixelCount(
        256,
        true
      );

    var smallPatches = classMask.and(

```

```

        connectedPixels.lt(
            minSnowAreaPixels
        )
    );

    cleaned = cleaned.where(
        smallPatches,
        localMajority
    );
});

return cleaned
    .rename('Snow_Class')
    .updateMask(
        inputClassImage.mask()
    )
    .toByte();
}

var classClean = applyMinimumMappingUnit(
    classImage
);

// Classification colours.
var classPalette = [
    '8c6d31', // 0 = Non-Snow: brown
    'ffffff', // 1 = Dry Snow: white
    '00bcd4' // 2 = Wet Snow: cyan
];

Map.addLayer(
    classClean,
    {
        min: 0,
        max: 2,
        palette: classPalette
    },
    'Snow classification: 0 Non-Snow, 1 Dry, 2 Wet',
    true
);

//
=====
====
// 9. AREA STATISTICS IN SQUARE KILOMETRES
//
=====
=====

```

```
function getClassAreaKm2(
  classValue
) {
```

```
  var result = ee.Image
    .pixelArea()
    .divide(1e6)
    .rename('area_km2')
    .updateMask(
      classClean.eq(
        classValue
      )
    )
    .reduceRegion({

      reducer:
        ee.Reducer.sum(),

      geometry:
        aoi,

      scale:
        processingScale,

      crs:
        exportCrs,

      maxPixels:
        maxPixels,

      tileScale:
        tileScale
    });
```

```
  return ee.Number(

    ee.Algorithms.If(

      result.contains(
        'area_km2'
      ),

      result.get(
        'area_km2'
      ),

      0
    )
  );
}
```

```
var nonSnowAreaKm2 = getClassAreaKm2(0);
```

```
var drySnowAreaKm2 = getClassAreaKm2(1);
```

```
var wetSnowAreaKm2 = getClassAreaKm2(2);
```



```
print(
  'Non-Snow area, km²:',
  nonSnowAreaKm2
);
```

```
print(
  'Dry Snow area, km²:',
  drySnowAreaKm2
);
```

```
print(
  'Wet Snow area, km²:',
  wetSnowAreaKm2
);
```

```
// Create one-row CSV summary.
```

```
var summaryTable = ee.FeatureCollection([
  ee.Feature(null, {
    year:
      analysisYear,

    season:
      analysisSeason,

    target_start:
      targetDates.start,

    target_end_exclusive:
      targetDates.end,

    sentinel2_image_count:
      sentinel2Collection.size(),

    sentinel1_target_image_count:
      sentinel1TargetCollection.size(),

    sentinel1_reference_image_count:
      sentinel1ReferenceCollection.size(),

    ndsi_snow_threshold:
      ndsiSnowThreshold,

    vv_absolute_threshold_db:
      vvAbsoluteThreshold,

    delta_vv_threshold_db:
      deltaVVThreshold,

    non_snow_area_km2:
      nonSnowAreaKm2,

    dry_snow_area_km2:
```

```

        drySnowAreaKm2,

        wet_snow_area_km2:
            wetSnowAreaKm2
    })
});

print(
    'Classification summary:',
    summaryTable
);

//
=====
====
// 10. LEGEND
//
=====
====

function addLegendRow(
    panel,
    color,
    label
) {

    var colorBox = ui.Label({

        style: {

            backgroundColor:
                '#' + color,

            padding:
                '8px',

            margin:
                '0 8px 4px 0'
        }
    });

    var textLabel = ui.Label({

        value:
            label,

        style: {

            margin:
                '0 0 4px 0'
        }
    });

    panel.add(

```

```

ui.Panel({

  widgets: [
    colorBox,
    textLabel
  ],

  layout:
    ui.Panel.Layout.Flow(
      'horizontal'
    )
  })
);
}

```

```

var legend = ui.Panel({

  style: {

    position:
      'bottom-left',

    padding:
      '8px 12px'
  }
});

```

```

legend.add(

  ui.Label({

    value:
      'Snow classification',

    style: {

      fontWeight:
        'bold',

      margin:
        '0 0 6px 0'
    }
  })
);

```

```

addLegendRow(
  legend,
  classPalette[0],
  'Non-Snow'
);

```

```

addLegendRow(
  legend,
  classPalette[1],
  'Dry Snow'

```

```

);

addLegendRow(
  legend,
  classPalette[2],
  'Wet Snow'
);

Map.add(legend);

//
=====
====
// 11. EXPORTS
//
// NoData values:
//
// Classification = 255
// Continuous images = -9999
//
=====
====

var exportPrefix =
  String(analysisYear) +
  '_' +
  analysisSeason;

// Classification export.

var classificationForExport = classClean
  .unmask({
    value: 255,
    sameFootprint: false
  });

Export.image.toDrive({

  image:
    classificationForExport.toUint8(),

  description:
    'Snow_Class_' + exportPrefix,

  folder:
    exportFolder,

  fileNamePrefix:
    'snow_class_' + exportPrefix,

  region:
    aoi,

  scale:

```

```

        processingScale,

        crs:
            exportCrs,

        maxPixels:
            maxPixels,

        fileFormat:
            'GeoTIFF',

        formatOptions: {

            noData:
                255,

            cloudOptimized:
                true
        }
    });

    // NDSI export.

    var ndsiForExport = ndsiImage
        .updateMask(validDataMask)
        .unmask({
            value: -9999,
            sameFootprint: false
        });

    Export.image.toDrive({

        image:
            ndsiForExport.toFloat(),

        description:
            'NDSI_' + exportPrefix,

        folder:
            exportFolder,

        fileNamePrefix:
            'ndsi_' + exportPrefix,

        region:
            aoi,

        scale:
            processingScale,

        crs:
            exportCrs,

        maxPixels:
            maxPixels,

        fileFormat:

```

```

        'GeoTIFF',

formatOptions: {

    noData:
        -9999,

    cloudOptimized:
        true
}
});

// Delta-VV export.

var deltaVVForExport = deltaVV
    .updateMask(validDataMask)
    .unmask({
        value: -9999,
        sameFootprint: false
    });

Export.image.toDrive({

    image:
        deltaVVForExport.toFloat(),

    description:
        'Delta_VV_' + exportPrefix,

    folder:
        exportFolder,

    fileNamePrefix:
        'delta_vv_' + exportPrefix,

    region:
        aoi,

    scale:
        processingScale,

    crs:
        exportCrs,

    maxPixels:
        maxPixels,

    fileFormat:
        'GeoTIFF',

    formatOptions: {

        noData:
            -9999,

        cloudOptimized:
            true
    }
});

```

```
}  
});
```

```
// Area-summary CSV export.
```

```
Export.table.toDrive({  
  
  collection:  
    summaryTable,  
  
  description:  
    'Snow_Class_Area_Summary_' +  
    exportPrefix,  
  
  folder:  
    exportFolder,  
  
  fileNamePrefix:  
    'snow_class_area_summary_' +  
    exportPrefix,  
  
  fileFormat:  
    'CSV'  
});
```

```
//
```

```
=====
```

```
=====  
// 12. FINAL METHOD STATEMENT  
//  
=====
```

```
=====
```

```
print(  
  
  'Applied classification rules:',  
  
  'Non-Snow: NDSI <= ' +  
    ndsiSnowThreshold +  
  
  '; Dry Snow: NDSI > ' +  
    ndsiSnowThreshold +  
    ' and neither wet-snow rule is met' +  
  
  '; Wet Snow: NDSI > ' +  
    ndsiSnowThreshold +  
  
  ' and (VV < ' +  
    vvAbsoluteThreshold +  
  
  ' dB OR deltaVV <= ' +  
    deltaVVThreshold +  
    ' dB).'  
);
```

```
print(  

```

'After checking the map, open the Tasks tab and run the export tasks.'
);